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REALIZATION CALCULUS



RESEARCH LIBRARY

Function Families and Their Structure

Asymmetric feedback, arithmetic depth, prime-sieve polynomials, and orbit spectra

Traditional paper

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REALIZATION CALCULUS

Reading scope

Four worked families. Definitions, first coefficients, growth laws, nonreduction results, and reproducible examples. Local proofs were independently reviewed. Historical novelty is unresolved for the candidate extensions.

This 1.0 release develops four families with full structural proofs, computed visuals, a reusable exact sequence API, and explicit polyphase and tap laws. Historical priority is classified separately from mathematical correctness.

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READING MAP

What was actually developed

The useful question is whether a definition carries structure that survives a change of notation. These four routes provide different answers: a scale law, an obstruction to finite differential equations, a detectable positivity boundary, and a rotation-invariant spectral coordinate.

Family	Structure worked out	Status of the construction
Asymmetric dilation	Entire solutions; exact growth balance; off-diagonal nonrescaling; intrinsic feedback banks	Candidate extension of a known one-dilation tangent analog
Complex arithmetic depth	Exact exponential type; coupled shift calculus; strict positive kernel; exact parameter recovery; no polynomial-coefficient linear ODE	Candidate extension of the project's integer arithmetic energy
Prime-sieve packets	Exact moment field; real Hankel singularities; positivity-margin collapse; multiple-prime geometry; arithmetic resolvent	Candidate arithmetic specialization of classical orthogonal-polynomial machinery
Orbit spectral zeta	Global meromorphic continuation; rational integer values; regularized product; multiplicity-aware orbit quotient	Worked specialization of known Mittag–Leffler seeds and spectral-zeta methods

A prior-art check changed the first route

The equation $f_q'(z) = 1 + f_q(qz)^2$, $f_q(0) = 0$, is already present in Miguel A. Méndez's 2024 path-length paper, Example 2.4. It remains an important baseline, but its definition is not a discovery of this project. The asymmetric extension keeps two separate contractions and permits a proved distinction from every amplitude/argument rescaling of that baseline.

How to read the evidence

Theorem statements below concern exact infinite objects under the displayed hypotheses. Plots and decimal roots illustrate them. Finite assertions check implementations within recorded ranges. None of these three forms of evidence establishes historical priority by itself.

The raw route files contain longer proofs, source-by-source comparisons, negative controls, and unresolved questions. Their exact revisions are pinned in the independent review and the final computation manifest.

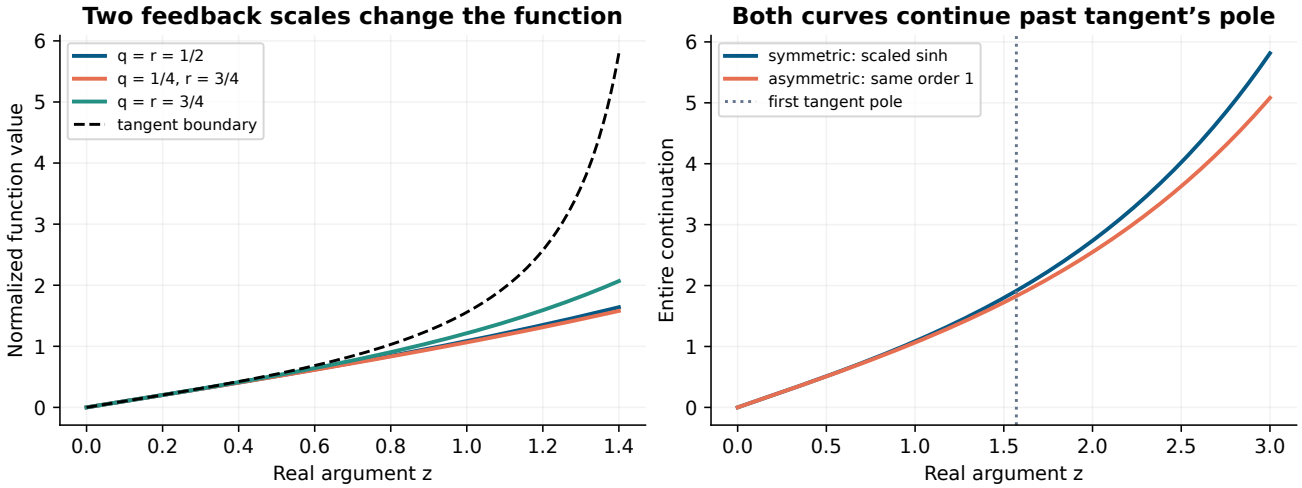
FAMILY 1

Two scales inside the function

For real $0 < q, r < 1$, define the normalized odd solution

$$\boxed{T'_{q,r}(z) = 1 + T_{q,r}(qz)T_{q,r}(rz), \qquad T_{q,r}(0) = 0.}$$

The contractions are part of the defining feedback. The function is symmetric in q, r, but unequal scales retain left/right address information.



Computed Taylor curves. Each entire curve uses terms through z^{321}; positive-coefficient Cauchy estimates bound the omitted tail on each displayed interval. The black dashed curve is ordinary tangent.

For $T_{q,r}(z) = \sum_{n\geq 0} c_n z^{2n+1}$, $c_0 = 1$, coefficient matching gives

$$c_n = \frac{1}{2n+1} \sum_{i=0}^{n-1} q^{2i+1} r^{2(n-1-i)+1} c_i c_{n-1-i}, \qquad n \geq 1.$$

In particular,

$$T_{q,r}(z) = z + \frac{qr}{3} z^3 + \frac{q^2 r^2 (q^2 + r^2)}{15} z^5 + O(z^7).$$

The coefficients are positive and bounded by tangent coefficients, giving a convergent germ. If the solution is known on $|z| < R$, its integral equation extends it to $|z| < R/\max(q, r)$. Repetition proves that it is entire.

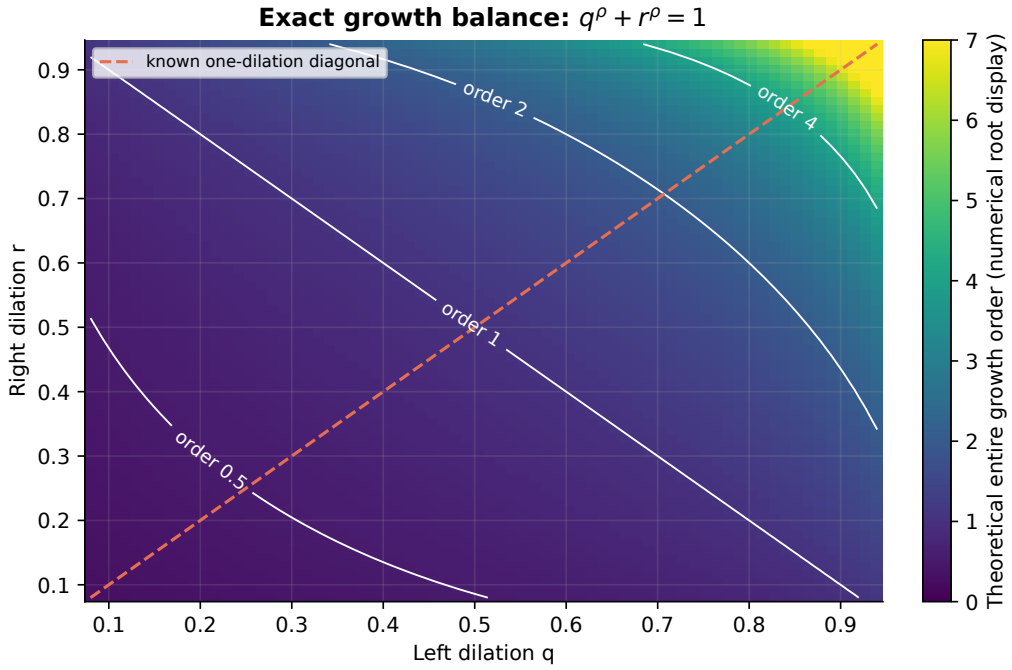
The parameters can be recovered, up to swapping, from c_1, c_2. Put $p = 3c_1$ and $u = 15c_2/p^2$. Then q, r are the roots of $x^2 - \sqrt{u + 2p}\, x + p = 0$.

FAMILY 1 / GROWTH

A scale-balance atlas

$$\boxed{\rho(T_{q,r}) = \rho \quad \text{where} \quad q^\rho + r^\rho = 1.}$$

Here $\rho(f) = \limsup_{R\to\infty} \log\log M_f(R)/\log R$ is entire-function order. Moreover $\log M_T(R) = O(R^\rho)$: the upper type is finite.



The displayed values numerically solve the proved characteristic equation. Colors saturate at order 7; the upper corner contains larger orders. The dashed diagonal is the previously known one-scale family.

Scales	Order	Interpretation
$q = r = 1/4$	$1/2$	Slow entire growth
$q + r = 1$	1	Many distinct functions at one order
$q = r = 1/\sqrt{2}$	2	Quadratic logarithmic growth scale
$q = r = 3/4$	$\log 2/\log(4/3)$	Approximately 2.4094208397

Why this equation appears

Nonnegative coefficients give $M(R) = T_{q,r}(R)$. The integral equation implies

$$M(R) \le R[1 + M(qR)M(rR)],$$

$$M(R) \ge (1-\alpha)RM(\alpha qR)M(\alpha rR), \quad 0 < \alpha < 1.$$

Taking logarithms turns product feedback into a sum over contracted scales. Comparison with CR^ρ proves the upper bound; comparison with cR^{ρ_α}, where $(\alpha q)^{\rho_\alpha} + (\alpha r)^{\rho_\alpha} = 1$, proves the lower bound. Letting $\alpha \uparrow 1$ gives the same order. A complete interval-induction proof is in the source route.

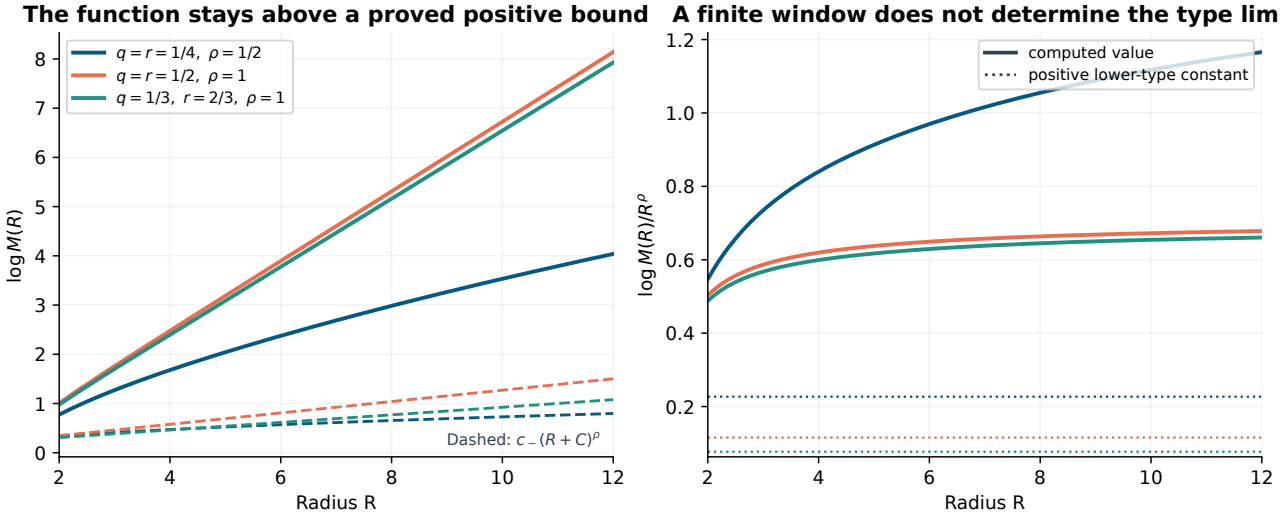
DILATION / 1.0 THEOREM

The exact order has positive lower type

The order theorem can now be strengthened. For every finite positive bank, with $\rho = \max_{h_j>0} \rho_j$,

$$0 < \liminf_{R\to\infty} \frac{\log M(R)}{R^\rho} \le \limsup_{R\to\infty} \frac{\log M(R)}{R^\rho} < \infty.$$

Thus logarithmic growth is trapped between two positive multiples of R^ρ. In particular $\log\log M(R)/\log R$ has a limit equal to ρ. The ratio $\log M(R)/R^\rho$ itself need not have a proved limit here.



Finite-radius logarithmic growth and its proved lower comparison. The normalized curves illustrate scale dependence; the plot is not a proof of convergence to a type constant. Analytic truncation envelopes and ordinary high-precision rounding are recorded separately.

The step that removes the old loss of exponent

Choose a maximizing channel (h, q, r) and integrate over a fixed-width terminal interval. Let

$$L = 1/h, \quad C = \max\left(\frac{qL}{1-q}, \frac{rL}{1-r}\right), \quad B = L + \frac{2}{\min(q,r)}, \quad c_- = \frac{\log 2}{(B+C)^\rho}.$$

The result is the explicit all-radius bound

$$\log M(R) \ge c_-(R+C)^\rho, \qquad R \ge 2.$$

Unlike a fixed fractional window, the terminal window loses only an additive radius. The shift C absorbs that loss without changing the exponent.

DILATION / PROOF

A translated power comparison

Write $H(R) = \log M(R)$. Positivity gives

$$M(R) \geq h \int_{R-L}^{R} M(qt)M(rt)\,dt \geq M(q(R-L))M(r(R-L)), \qquad R \geq B.$$

The choice of C ensures

$$q(R-L)+C \geq q(R+C), \qquad r(R-L)+C \geq r(R+C).$$

On the seed interval $[2, B]$, $H(R) \geq \log R \geq \log 2 \geq c_-(R+C)^\rho$. If this comparison holds at the two child radii, then

$$\begin{aligned} H(R) &\geq H(q(R-L)) + H(r(R-L)) \\ &\geq c_-\{[q(R-L)+C]^\rho + [r(R-L)+C]^\rho\} \\ &\geq c_-(q^\rho + r^\rho)(R+C)^\rho = c_-(R+C)^\rho. \end{aligned}$$

Let $Q_* = \max(q, r)$ and $d = (1-Q_*)B + Q_*L > 0$. Each child radius is at most $R-d$ and at least two. Induction on successive intervals of length d propagates the seed comparison to every $R \geq B$. The argument uses monotonicity of positive powers and works also when $\rho < 1$.

Explicit upper constants are available too

Set $Q = \max_j \max(q_j, r_j)$, $m = \min_j \min(q_j, r_j)$, $\Delta = \log(1/m)$, and $B_+ = \max_j \log(1/(q_j r_j)) + \log 2$. Choose an integer K with $Q^{-K} \geq 3/m$ and put

$$V_K = (2^{K+1}-1)\log 2 + (2^{K+1}-K-2)\log(1/Q),$$

$$c_+ = e^{-\rho}(V_K + 1 + \Delta + B_+).$$

Then $\log M(R) \leq c_+ R^\rho - \log R - B_+$ for $R \geq e$. The tangent majorant $M(1) < 2$, followed by the integral recurrence on radii Q^{-k}, bounds the initial seed by V_K. The logarithmic-radius comparison then supplies the displayed upper constant. The full induction is in Appendix A.

These constants are constructive and conservative. Their purpose is to establish two-sided growth, not to estimate the optimal type.

FAMILY 1 / SEPARATION

More than a rescaled one-scale seed

Nonrescaling theorem

If $q \neq r$, then $T_{q,r}$ cannot equal $Af_s(Bz)$ for nonzero constants A, B and any real $0 < s < 1$, where $f_s' = 1 + f_s(sz)^2$ is the known family.

Suppose such an equality held. Normalization gives $AB = 1$. Write $p = qr$, $u = q^2 + r^2$, and let ρ be the common order. The first two nontrivial coefficients and the growth law force

$$B^2 s^2 = p, \qquad s^2 = u/2, \qquad 2s^\rho = q^\rho + r^\rho = 1.$$

Consequently

$$2\left(\frac{q^2 + r^2}{2}\right)^{\rho/2} = q^\rho + r^\rho.$$

For $\rho \neq 2$, strict convexity or concavity of $x^{\rho/2}$ forces $q = r$. For the exceptional case $\rho = 2$, we have $u = 1$, $s^2 = 1/2$, $B^2 = 2p$. The two z^7 coefficients become

$$\frac{p^3(3 - p^2)}{315} \quad \text{and} \quad \frac{11p^3}{1260}.$$

Their equality forces $p = 1/2$ and again $(q - r)^2 = u - 2p = 0$. This handles the exceptional order rather than assuming it away.

A worked order-one member

$$T_{1/3,2/3}(z) = z + \frac{2}{27}z^3 + \frac{4}{2187}z^5 + \frac{232}{11160261}z^7 + O(z^9).$$

It has order one, and $T_{1/3,2/3}(1) \approx 1.0759239897408249$. The diagonal point $q = r = 1/2$ gives $f_{1/2}(z) = \sqrt{2}\sinh(z/\sqrt{2})$. Every off-diagonal member is separated from all rescaled diagonal members by the theorem above.

A combinatorial address formula

Sum over full ordered binary trees with n internal vertices. A vertex with i internal vertices in its left subtree and j in its right contributes

$$\frac{q^{2i+1}r^{2j+1}}{2(i + j + 1) + 1}.$$

Multiply these factors over internal vertices, then sum over trees. Root splitting reproduces the coefficient recurrence exactly. Unequal contractions therefore record left and right path lengths separately.

This is a precise separation result from one authenticated antecedent. It does not prove that no earlier author studied the asymmetric equation.

FAMILY 1 / TAPS

Feedback banks and weak-channel sensitivity

Choose finitely many channels $j = 1, \dots, J$, with $h_j > 0$, $\sum_j h_j = 1$, and $0 < q_j, r_j < 1$. Define

$$F'_\Theta(z) = 1 + \sum_j h_j F_\Theta(q_j z) F_\Theta(r_j z), \qquad F_\Theta(0) = 0.$$

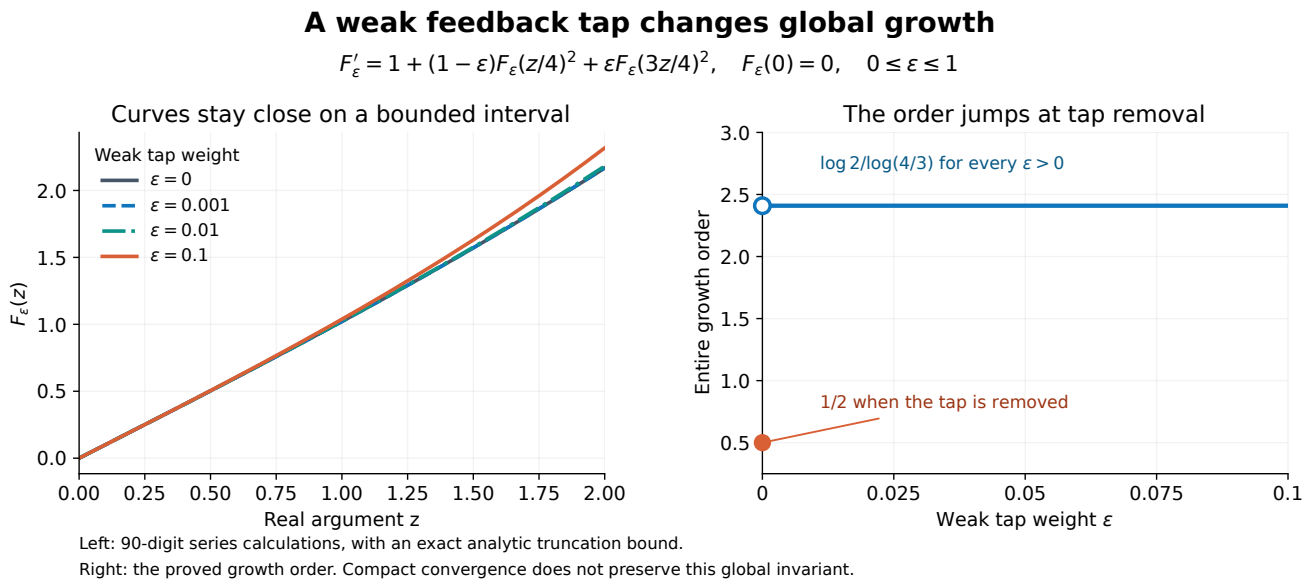
These taps act inside the nonlinear definition. With $\mu_1 = \sum_j h_j q_j r_j$ and $\mu_2 = \sum_j h_j q_j r_j (q_j^2 + r_j^2)$,

$$F_\Theta(z) = z + \frac{\mu_1}{3} z^3 + \frac{\mu_1 \mu_2}{15} z^5 + O(z^7).$$

If $q_j^{\rho_j} + r_j^{\rho_j} = 1$, then

$$\boxed{\rho(F_\Theta) = \max_{j \text{ active}} \rho_j.}$$

The positive weights alter local coefficients; the most expansive active channel determines the asymptotic order. A zero-weight channel is removed before taking the maximum.



Left: 90-digit series calculations, with an exact analytic truncation bound.
Right: the proved growth order. Compact convergence does not preserve this global invariant.

A two-channel example: scales $(1/4, 1/4)$ with weight $1 - \varepsilon$, and $(3/4, 3/4)$ with weight ε. The function varies continuously on every fixed compact set, while its growth order jumps when the second channel becomes active.

For $0 < \varepsilon \le 1$ the order is $\log 2 / \log(4/3)$; at $\varepsilon = 0$ it is $1/2$. Coefficients depend continuously on ε and are dominated by those of the entire $f_{3/4}$. Dominated convergence of the Taylor series therefore proves uniform convergence on compact sets as $\varepsilon \downarrow 0$.

This gives a useful design distinction: finite-window similarity does not control entire growth order. For a bank with weights $1/4, 3/4$ at these scales,

$$F(z) = z + \frac{7}{48} z^3 + \frac{427}{30720} z^5 + \frac{176681}{188743680} z^7 + O(z^9).$$

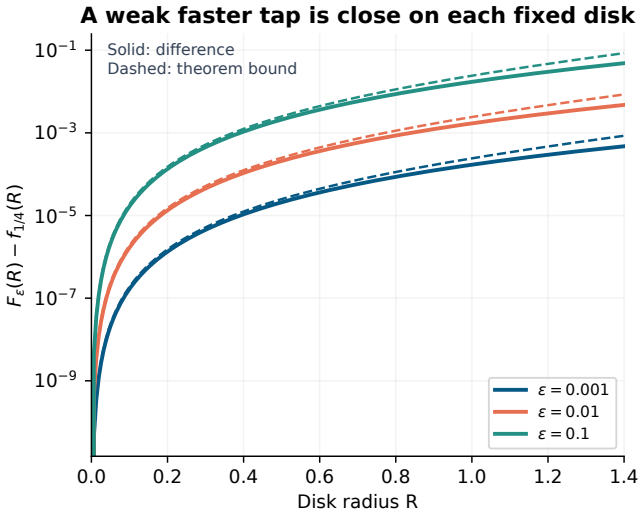
DILATION / TAPS

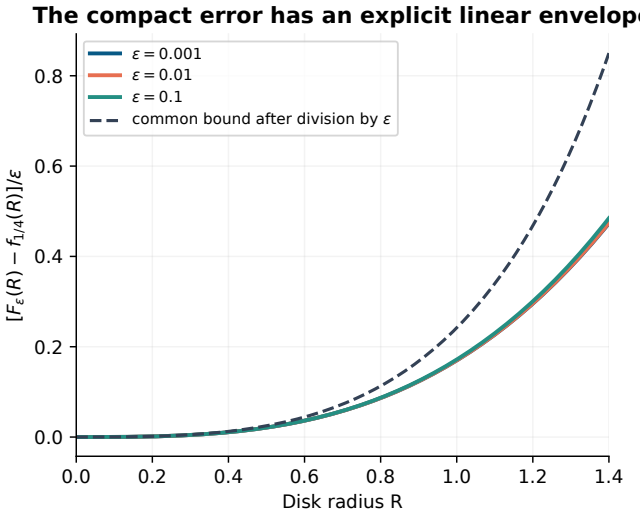
A quantitative compact-continuity law

Fix the contractions and let h, g be two normalized nonnegative tap vectors. Set $Q = \max_j \max(q_j, r_j) < 1$ and $d = \|h - g\|_1$. Then

$$\sup_{|z|\le R} |F_h(z) - F_g(z)| \le \frac{d}{4}\,[Rf_Q'(R) - f_Q(R)].$$

The right side is an explicitly defined entire majorant. It measures local sensitivity without asserting continuity of the asymptotic growth order.





Computed weak-tap differences lie below the proved compact bound. For a slow/faster two-channel bank, $d = 2\varepsilon$, so the envelope is $\varepsilon[Rf_{3/4}'(R) - f_{3/4}(R)]/2$.

Each $c_n(h)$ is a positive polynomial homogeneous of degree n in the weights. Coefficientwise comparison gives $c_n(h) \le d_n(Q)(\sum_j h_j)^n$, where $f_Q = \sum_n d_n(Q) z^{2n+1}$. Every weight partial derivative on the simplex lies in $[0, nd_n(Q)]$. The positive and negative parts of $h - g$ each have mass $d/2$, giving $|c_n(h) - c_n(g)| \le (d/2)nd_n(Q)$. Summing this inequality uses $\sum_n nd_n(Q)R^{2n+1} = [Rf_Q'(R) - f_Q(R)]/2$.

DILATION / PARAMETERS

An analytic family with a discontinuous invariant

For fixed finite J, allow complex h_j, q_j, r_j, with $|q_j|, |r_j| < 1$. The coefficient recurrence defines polynomials in all these parameters. On a compact parameter set choose $Q < 1$ and $W > 0$ such that $|q_j|, |r_j| \le Q$ and $\sum_j |h_j| \le W$. Then

$$|c_n(h,q,r)| \le W^n d_n(Q), \qquad \sum_n |c_n| R^{2n+1} \le W^{-1/2} f_Q(\sqrt{W}R).$$

Normal convergence proves joint holomorphic dependence on parameters and argument throughout this domain. Signed and complex banks have entire solutions, but the positive-bank growth theorem is not asserted for them.

The weak-tap example now has both sides quantified

For the bank with scales $(1/4, 1/4)$ and $(3/4, 3/4)$, weights $1-\varepsilon, \varepsilon$, the functions converge on every compact set as $\varepsilon \downarrow 0$. The preceding envelope gives a linear bound in ε. At every $\varepsilon > 0$, the order is $\log 2 / \log(4/3)$, while at zero it is $1/2$.

The new lower-type theorem does not contradict this: its lower constant depends on the active faster weight and can tend to zero as that channel vanishes. For $\varepsilon = 10^{-3}$ one constructive lower constant is about 1.4496×10^{-9}; a conservative upper constant is about 90.4392.

Identifying a bank has unavoidable equivalences

Permuting channels, swapping the scales within a channel, or merging duplicate scale pairs by adding their weights leaves the function unchanged. The two-scale single-channel inverse is exact up to swapping. A general minimal-bank identifiability theorem beyond these equivalences is not proved.

The resulting research target is specific: determine optimal growth type, possible logarithmic oscillations, and minimal-bank identifiability. A finite plot cannot settle any of those questions.

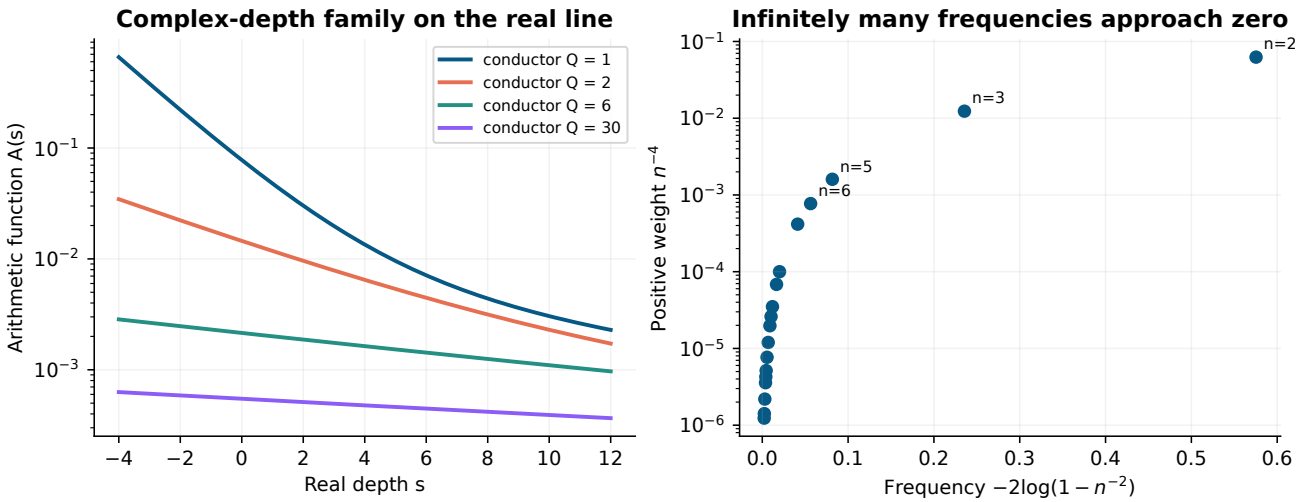
FAMILY 2

Arithmetic energy at complex depth

For real $a > 1/2$, $\beta > 0$, and integer $Q \geq 1$, define

$$A_{a,\beta,Q}(s) = \sum_{\substack{n \geq 2 \\ (n,Q)=1}} \mu(n)^2 n^{-2a} (1 - n^{-\beta})^{2s}, \qquad s \in \mathbb{C}.$$

The complex power uses the real logarithm of $1 - n^{-\beta} > 0$. The atom $n = 1$ is excluded because it has no such logarithm. Thus positive integer depths match the inherited energy, while $A(0) = E(0) - 1$.



Left: $a = \beta = 2$ at real depths, using the exact zeta–Newton representation at 90-digit working precision. Right: initial squarefree frequencies and weights for $Q = 1$. Only finitely many atoms are shown; the proof uses the infinite support.

Set $\lambda_n = -2\log(1 - n^{-\beta})$. Then $A(s) = \sum w_n e^{-\lambda_n s}$, $w_n = \mu(n)^2 n^{-2a}$. The weights are summable, and the distinct frequencies decrease to zero. If p_* is the least prime not dividing Q, then

$$\tau = -2\log(1 - p_*^{-\beta}), \qquad M = Z_Q(2a) - 1,$$

$$|A(s)| \leq M e^{\tau \max(-\Re s, 0)}.$$

This proves local uniform convergence and entire continuation. The first atom dominates $A(-t)$ as $t \to \infty$, so the entire function has exact order one and exponential type τ.

FAMILY 2 / CALCULUS

Closed shifts, infinite differential complexity

Write

$$Z_Q(v) = \frac{\zeta(v)}{\zeta(2v)} \prod_{p|Q} (1 + p^{-v})^{-1}, \qquad v > 1.$$

Only $\mathrm{rad}(Q)$ matters. Expansion of the depth factor gives

$$\boxed{A_{a,\beta,Q}(s+1) = A_{a,\beta,Q}(s) - 2A_{a+\beta/2,\beta,Q}(s) + A_{a+\beta,\beta,Q}(s).}$$

The fractional transfer and Newton formulas are

$$A_{a,\beta,Q}(s+t) = \sum_{j\geq 0} (-1)^j \binom{2t}{j} A_{a+j\beta/2,\beta,Q}(s),$$

$$A_{a,\beta,Q}(t) = \sum_{j\geq 0} (-1)^j \binom{2t}{j} [Z_Q(2a + j\beta) - 1].$$

They converge absolutely and locally uniformly: binomial coefficients have at most polynomial growth on compact parameter sets, while the shifted sums decay geometrically. The subtraction of one is essential at noninteger depth.

No polynomial-coefficient linear ODE

Suppose $\sum_{j=0}^{r} P_j(s) A^{(j)}(s) = 0$ with polynomial P_j. At $s = -t$, divide by the exponential of the first support atom. The remaining tail is exponentially smaller, so the polynomial $\sum_j P_j(-t)(-\lambda_{p_*})^j$ tends to zero exponentially and must vanish identically. Remove that atom and repeat. Every support frequency therefore satisfies $\sum_j P_j(s)(-\lambda_n)^j = 0$ for all s. For fixed s, this polynomial in λ has infinitely many distinct roots. All its coefficients vanish. Thus all P_j are zero, a contradiction.

The function is **not D-finite in depth**. This excludes identification in the depth variable with fixed finite-parameter hypergeometric or Meijer G functions and ordinary finite D-finite combinations. It does not exclude nonalgebraic substitutions or representations by known non-D-finite functions.

Strict positive complex kernel

For distinct complex points s_i, $K_{ij} = A(s_i + \overline{s_j})$ is strictly positive definite. A zero quadratic form would force a finite exponential sum to vanish at every λ_n. Their accumulation at zero invokes the identity theorem; derivatives at zero then give a nonsingular Vandermonde system. On the real axis, $(-1)^j A^{(j)}(s) > 0$ for every $j \geq 0$.

FAMILY 2 / ASYMPTOTICS

An arithmetic sector law

The negative real direction is governed by the first admissible prime. The positive direction is governed by the density of infinitely many squarefree integers. These are different pieces of the same entire function.

Let

$$\alpha = \frac{2a-1}{\beta}, \qquad \delta_Q = \frac{1}{\zeta(2)} \prod_{p \mid Q} \frac{p}{p+1}.$$

As $|s| \to \infty$ in any closed sector $|\arg s| \le \pi/2 - \epsilon$ with $\epsilon > 0$,

$$\boxed{A_{a,\beta,Q}(s) = \frac{\delta_Q}{\beta} B(\alpha, 2s+1) + O_{a,\beta,Q,\epsilon}\left(|s|^{-(2a-1/2)/\beta}\right).}$$

In particular,

$$A_{a,\beta,Q}(s) \sim \frac{\delta_Q \Gamma(\alpha)}{\beta} (2s)^{-\alpha}.$$

The principal power is used in the right half-plane. The beta expression is an asymptotic continuous-density model, not an identity with the arithmetic sum.

Proof mechanism

Squarefree inclusion–exclusion gives

$$\sum_{\substack{n \le x \\ (n,Q)=1}} \mu(n)^2 = \delta_Q x + O_Q(\sqrt{x}).$$

Integrate against $g_s(x) = x^{-2a}(1 - x^{-\beta})^{2s}$ on $[1, \infty)$. The density term becomes the beta integral after $y = x^{-\beta}$. Partial integration bounds the error by $C_Q \int_1^\infty \sqrt{x} |g_s'(x)|\, dx$. The two derivative terms are beta integrals with decay exponent $(2a - 1/2)/\beta$; $|s|/\Re s$ stays bounded in the stated sector. This gives the error term and then the leading asymptotic.

One complex example

At $a = \beta = 2$, $Q = 1$,

$$A(1/2 + i/4) \approx 0.06033183289343401791658 - 0.00741238405378543625 1981\, i.$$

Direct finite sums and a zeta–Newton evaluation agree within the recorded omitted-tail envelope. This is an ordinary high-precision example, not an interval certificate.

Mixed Dirichlet-character weights provide matrix-valued positive kernels. Individual off-diagonal entries may be signed or complex, so complete monotonicity is asserted only for the positive principal family.

ARITHMETIC DEPTH / INVERSE

The function encodes its parameters

The complete exact analytic germ determines

$$\boxed{a, \quad \beta, \quad \operatorname{rad}(Q).}$$

Precisely, two admissible triples define the same germ if and only if these three quantities agree. Prime powers in Q are invisible because the terms use only coprimality. A finite Taylor polynomial or noisy measurements are different inputs from the complete germ in this theorem.

Let the support integers be $n_1 < n_2 < \cdots$, frequencies $\lambda_j = -2\log(1 - n_j^{-\beta})$, and weights $w_j = n_j^{-2a}$. Starting at $R_1 = A$, peel the ordered atoms:

$$\lambda_j = \lim_{t\to\infty} \frac{\log R_j(-t)}{t}, \qquad w_j = \lim_{t\to\infty} e^{-\lambda_j t} R_j(-t),$$

$$R_{j+1}(s) = R_j(s) - w_j e^{-\lambda_j s}.$$

Every limit follows from the isolated largest remaining frequency and the positive summable tail. Thus the exact function determines its spectral list.

The counting law $N_Q(x) = \delta_Q x + O_Q(\sqrt{x})$ gives

$$\lambda_j = 2\delta_Q^{\beta} j^{-\beta}[1 + O_{Q,\beta}(j^{-1/2} + j^{-\beta})], \quad \beta = -\lim_{j\to\infty} \frac{\log \lambda_j}{\log j}.$$

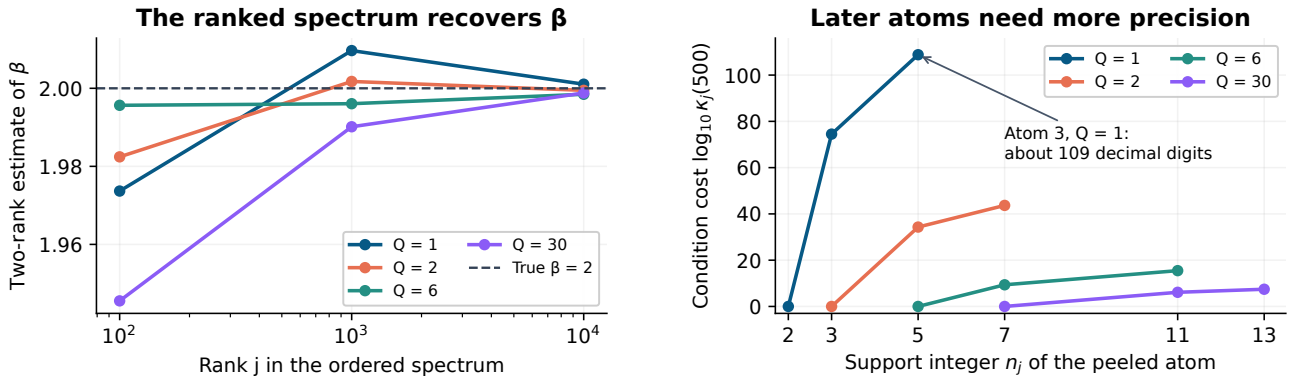
Once β is recovered,

$$n_j = (1 - e^{-\lambda_j/2})^{-1/\beta}, \qquad a = -\frac{\log w_j}{2\log n_j}.$$

Prime membership in this integer support then recovers the excluded prime set. The reconstruction is exact from infinite data; it does not certify that a finite search has found every excluded prime.

ARITHMETIC DEPTH / RESOLUTION

Separation improves while subtraction worsens



a = β = 2 · negative depth t = 500 · previous atoms removed exactly

Checked high-precision examples. Exact recovery uses the full analytic germ.

The frequency-rank estimates converge toward the depth exponent. Later-atom peeling at large negative depth also has a rapidly growing subtraction condition factor. These are finite examples from the checked arithmetic run, not certified floating-point intervals.

If earlier atoms have been removed exactly, set $\Delta_j = \lambda_j - \lambda_{j+1} > 0$, $M_{j+1} = \sum_{k>j} w_k$. The logarithmic secant

$$\widehat{\lambda}_j(t,h) = \frac{\log R_j(-t-h) - \log R_j(-t)}{h}$$

obeys, for $t \geq 0, h > 0$,

$$0 \leq \lambda_j - \widehat{\lambda}_j(t,h) \leq \frac{1}{h}\log\left(1 + \frac{M_{j+1}}{w_j}e^{-\Delta_j t}\right).$$

This improves with negative depth. But relative error in $A(-t)$ can be amplified after subtraction by

$$\kappa_j(t) = \frac{A(-t)}{R_j(-t)} \sim \frac{w_1}{w_j}e^{(\lambda_1-\lambda_j)t}.$$

For the displayed $Q = 1$ example, the third-atom subtraction at $t = 500$ has a condition factor of roughly $10^{108.8}$. Exact uniqueness therefore does not promise easy recovery from ordinary precision.

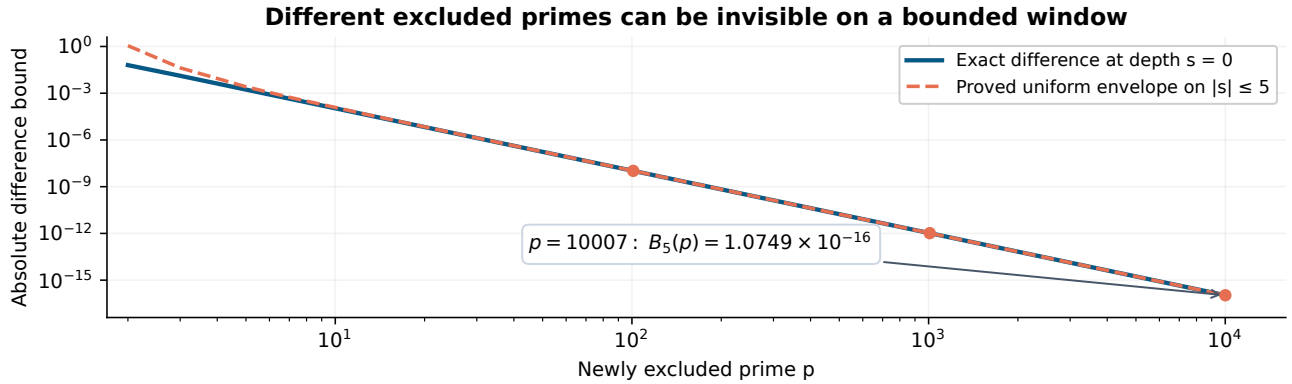
ARITHMETIC DEPTH / STABILITY

A large prime can hide in every finite window

For a prime $p \nmid Q$, any $R \geq 0$, and integer $k \geq 0$,

$$\sup_{|s|\leq R} |A^{(k)}_{a,\beta,Q}(s) - A^{(k)}_{a,\beta,Qp}(s)| \leq \lambda_p^k e^{R\lambda_p} \frac{Z_Q(2a)}{p^{2a}+1}.$$

The difference consists of the p-divisible admissible squarefree atoms. Their frequencies are at most λ_p and their total weight is the Euler-factor expression on the right. Termwise differentiation proves the bound for every derivative.



Exact theorem envelopes evaluated at 90 decimal digits; numerical display is not ball-certified.

The proved compact-window envelope shrinks as the excluded prime increases. At $a = \beta = 2$, $Q = 1$, $R = 5$, excluding $p = 10007$ changes the entire window by at most approximately 1.0749×10^{-16}. The depth-zero difference is known exactly.

The radical differs for every such prime, while all bounded-window derivatives converge to those of A_Q. Without a bound on excluded primes, there is no uniformly stable inverse from these windows to the radical parameter.

The same distinction appears in weak taps: exact structural invariants can change under perturbations that are small at finite resolution.

ARITHMETIC DEPTH / REALIZATION

An infinite feature space with arithmetic labels

For the ordered support, define $\phi_s(j) = \sqrt{w_j} e^{-\lambda_j s}$. The kernel $A(s+\bar{t})$ is its weighted ℓ^2 Gram kernel. Summability and bounded frequencies make every feature vector well defined.

If a weighted ℓ^2 combination produces the zero entire function, Cauchy–Schwarz makes its coefficients absolutely summable. Successive dominant-frequency peeling then forces every coefficient to vanish. This gives an injective Hilbert realization, and all finite derivative Hankel matrices have full rank: a nonzero polynomial in λ cannot vanish on the infinite frequency support.

Fixed-modulus character resolution

For characters χ, ψ modulo a fixed Q, let $\eta = \chi\overline{\psi}$. When η is nonprincipal, the elementary squarefree character sum is $O_Q(\sqrt{x})$, with no density term. The same sector partial-integration argument therefore gives

$$A_{\chi,\psi}(s) = O_{a,\beta,Q,\epsilon}\big(|s|^{-(2a-1/2)/\beta}\big), \qquad |\arg s| \le \pi/2 - \epsilon.$$

Compared with the principal $|s|^{-(2a-1)/\beta}$ scale, this is faster by $|s|^{-1/(2\beta)}$. The modulus is fixed; the constants are not asserted uniform as Q grows. Individual off-diagonal functions can be complex and are not asserted completely monotone.

Classical Laplace uniqueness provides a comparison for the inverse mechanism. The parameter-specific extraction, frequency-rank law, and explicit instability are proved locally in the complete arithmetic route.

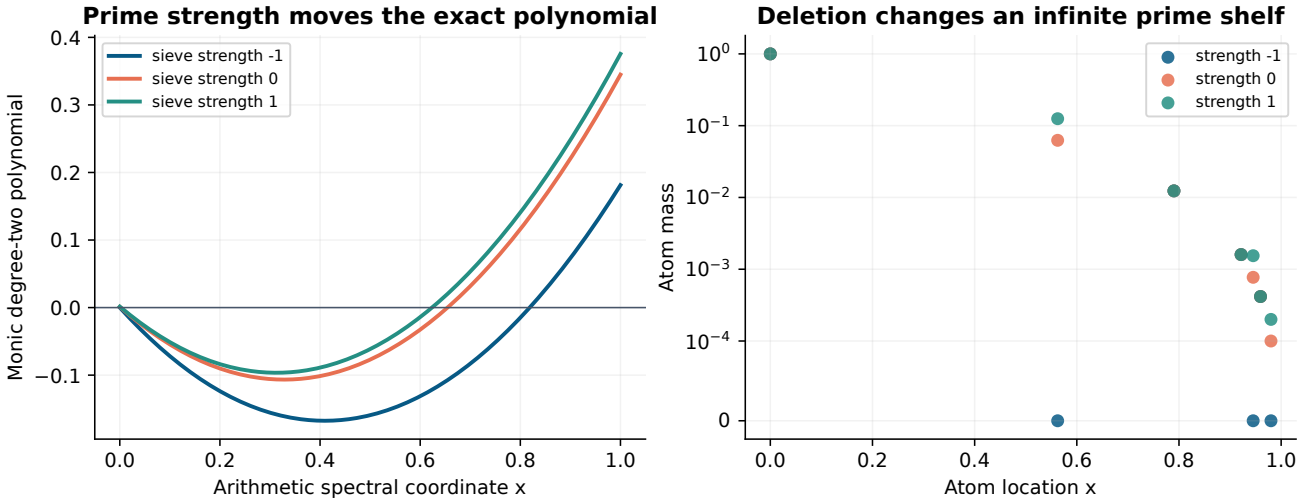
FAMILY 3

An infinite prime shelf changes the packets

For a prime p and $\lambda \geq -1$, let

$$x_n = (1 - n^{-2})^2, \qquad \nu_{p,\lambda} = \sum_{n \geq 1} \mu(n)^2 n^{-4} (1 + \lambda \mathbf{1}_{p|n}) \delta_{x_n}.$$

Define monic orthogonal polynomials $P_m^{(k;p,\lambda)}$ for the measure $x^k \, d\nu_{p,\lambda}$, $k \geq 0$ integer. The zero atom has mass one at $k = 0$ and is annihilated when $k \geq 1$.



Left: degree-two packets for $p = 2$, depth zero. Right: initial atom masses. Strength -1 deletes every even squarefree atom; strength 0 is the original measure; strength 1 doubles that shelf.

The exact moments are finite zeta stencils:

$$M_j^{p,\lambda} = \sum_{r=0}^{2j} (-1)^r \binom{2j}{r} \frac{\zeta(4+2r)}{\zeta(8+4r)} \left(1 + \frac{\lambda}{p^{4+2r}+1}\right).$$

All polynomial coefficients belong to $\mathbb{Q}(\pi^{-2}, \lambda)$. With $\Delta = M_0 M_2 - M_1^2$,

$$P_1 = x - \frac{M_1}{M_0}, \qquad P_2 = x^2 - \frac{M_0 M_3 - M_1 M_2}{\Delta} x + \frac{M_1 M_3 - M_2^2}{\Delta}.$$

The infinitely many positive nodes make every Hankel matrix strictly positive definite throughout $\lambda \geq -1$. Hence all degrees exist, with positive norms, real simple zeros in $(0, 1)$, and interlacing.

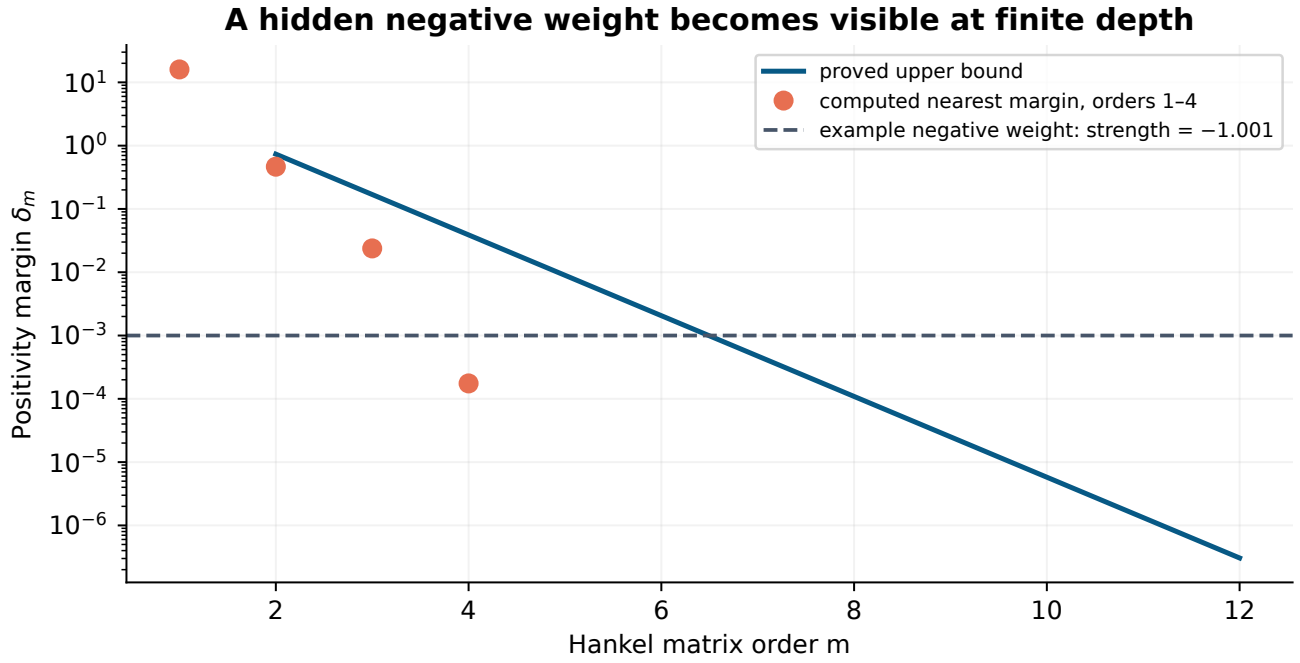
FAMILY 3 / BOUNDARY

Finite positivity can conceal a signed defect

Split $\nu_{p,\lambda} = \nu_{\mathrm{non}p} + (1+\lambda)\nu_p$. Let A_m, B_m be their order-m Hankel matrices. Both are positive definite. Thus

$$\tau_m(\lambda) = \det(A_m + (1+\lambda)B_m) = \det A_m \prod_{i=1}^{m}[1 + (1+\lambda)\eta_i],$$

where $\eta_i > 0$ are the eigenvalues of $A_m^{-1/2} B_m A_m^{-1/2}$. Every determinant root is real and below -1; possible coefficient poles occur there, although numerator cancellation is possible.



The line is a proved bound for $p = 2$; dots are ordinary high-precision pencil roots. A defect $\lambda = -1.001$ eventually violates positivity even though it can pass low-order checks.

The nearest root is $-1 - \delta_m$, with

$$\delta_m = \min_{0 \neq f,\, \deg f < m} \frac{\int f^2 \, d\nu_{\mathrm{non}p}}{\int f^2 \, d\nu_p} > 0, \qquad \delta_m \downarrow 0.$$

To prove the limit, approximate a continuous bump supported at the isolated node x_p by polynomials on $[0,1]$. The numerator tends to zero while the denominator retains the p^{-4} atom. Hence every $\lambda < -1$ eventually produces a negative finite-degree quadratic form.

For $p = 2$, test with $f(x) = x(1-x)^{m-2}$, $m \geq 2$:

$$\delta_m \leq \frac{4096}{81}\left(\frac{16M_0}{17} - 1\right)\left(\frac{272}{567}\right)^{2(m-2)}, \qquad M_0 = 105\pi^{-4}.$$

This follows from $x_2 = 9/16$ and $x_n \geq 64/81$ at nonzero odd nodes. Exactly $\delta_1 = p^4$; the limit is an infinite-resolution phenomenon.

FAMILY 3 / TRANSPORT

Classical operations with arithmetic coordinates

The depth tilt x^k is a Christoffel transformation. Suppressing p, λ,

$$P_m^{(k+1)}(x) = \frac{P_{m+1}^{(k)}(x) - [P_{m+1}^{(k)}(0)/P_m^{(k)}(0)]P_m^{(k)}(x)}{x}.$$

The numerator vanishes at zero; all roots are positive, so the displayed denominator coefficient is nonzero. For shifted Hankel determinants $\tau_m^{(k)} = \det[M_{k+i+j}]_{i,j<m}$, with $\tau_0^{(k)} = 1$,

$$\tau_m^{(k)}\tau_m^{(k+2)} - (\tau_m^{(k+1)})^2 = \tau_{m+1}^{(k)}\tau_{m-1}^{(k+2)} > 0, \qquad m \geq 1.$$

This is the Desnanot–Jacobi identity applied to exact arithmetic moments. The construction mechanism is classical.

An endpoint mass is numerically delicate

For the undeformed depth-zero measure, set $\nu_\theta = \nu + \theta\delta_0$, $\theta \geq -1$. If $K_m(x,y) = \sum_{j=0}^m P_j(x)P_j(y)/h_j$, the Uvarov formula is

$$P_m^{[\theta]}(x) = P_m(x) - \frac{\theta P_m(0)}{1 + \theta K_{m-1}(0,0)} K_{m-1}(x,0),$$

with $K_{-1} = 0$ for degree zero. The mass at zero is recovered by

$$1 \leq \frac{1}{K_m(0,0)} \leq 1 + (M_0 - 1)(7/16)^{2m}.$$

The lower bound comes from the mass-one zero atom. The test polynomial $(1-x)^m$ gives the upper bound since every other node is at least $9/16$. At finite degree $K_m(0,0) < 1$, because no nonzero polynomial can vanish at all the other infinitely many nodes. Complete mass removal is valid, while $1 - K_m(0,0)$ becomes very small. This explains the conditioning problem.

The resolvent is non-D-finite

The Stieltjes function $S_{p,\lambda}(z) = \int (z-x)^{-1}\, d\nu_{p,\lambda}(x)$ has infinitely many isolated poles at the positive-weight nodes $x_n \to 1$. A solution of a finite rational-coefficient linear ODE has only finitely many possible finite singular points. The germ at $z = 2$ therefore is not D-finite. This is a general infinite-atomic-resolvent mechanism, specialized here to arithmetic data; it is not a priority claim.

The full packet sequence also cannot be an affine copy of a continuous Jacobi sequence: compact moment determinacy would equate a purely atomic measure with a continuous one. An inherited Laguerre scaling limit remains compatible with this separation of the complete families.

PRIME SIEVES / MULTIPLE PARAMETERS

Overlap is part of the arithmetic

For a finite set F of distinct primes, define

$$\nu_{F,\lambda} = \sum_{n\geq 1} \mu(n)^2 n^{-4} \prod_{p\in F}(1+\lambda_p \mathbf{1}_{p|n})\delta_{x_n}, \quad x_n = (1-n^{-2})^2.$$

For real strengths, the exact positivity domain is

$$\boxed{\lambda_p \geq -1 \quad \text{for every } p \in F.}$$

Sufficiency is termwise positivity. If one strength is below -1, the atom $n = p$ isolates its negative factor. The other strengths do not occur at that atom. At every simultaneous deletion endpoint, primes outside F still supply infinitely many positive nodes, so all Hankel matrices remain strictly positive definite.

The exact coordinate formula is

$$M_j(\lambda) = \sum_{r=0}^{2j} (-1)^r \binom{2j}{r} \frac{\zeta(4+2r)}{\zeta(8+4r)} \prod_{p\in F}\left(1+\frac{\lambda_p}{p^{4+2r}+1}\right).$$

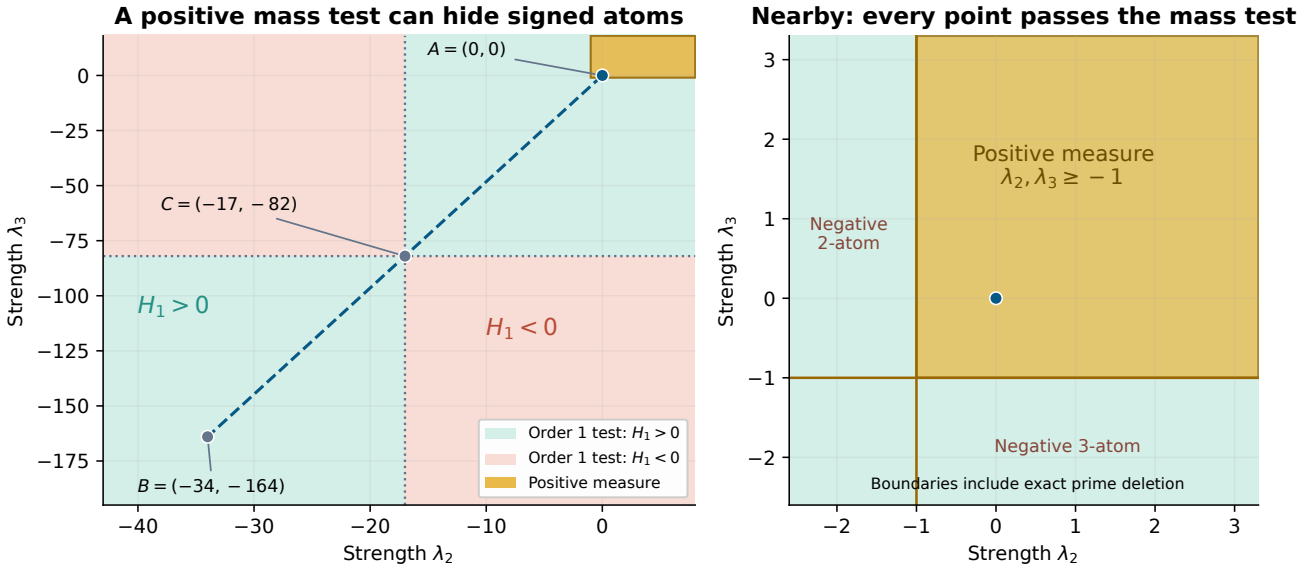
Coefficients of the monic families lie in $\mathbb{Q}(\pi^{-2}, \lambda_p : p \in F)$. For two primes, the overlap term is visible:

$$M_j = M_j(0,0) + \lambda_p U_{p,j} + \lambda_q U_{q,j} + \lambda_p \lambda_q U_{pq,j}.$$

Here $U_{pq,j} > 0$ is the pq-divisible submeasure moment. The family does not reduce to adding separate prime responses without their overlap.

PRIME SIEVES / FINITE TESTS

Moment positivity is a weaker requirement



The exact positive-measure domain and the order-one moment test have different parameter geometry. Regions passing a finite test outside the safe orthant still contain negative atoms.

For primes two and three,

$$M_0 = 105\pi^{-4}(1 + \lambda_2/17)(1 + \lambda_3/82).$$

Both $(0, 0)$ and $(-34, -164)$ have positive total mass. Their midpoint $(-17, -82)$ has zero mass. Thus the parameter set passing the order-one positive-definiteness test is nonconvex. This particular midpoint argument does not establish nonconvexity of the semidefinite test, which permits zero.

A separate additive family has different geometry

For weights $1 + \sum_{p \in F} \gamma_p \mathbf{1}_{p|n}$, positivity is equivalent to

$$1 + \sum_{\gamma_p < 0} \gamma_p \geq 0.$$

Every prime subset appears at its squarefree product, proving necessity. Checking each atom proves sufficiency. Its finite Hankel matrices are affine in γ, so the semidefinite feasible sets are convex spectrahedra. The multiplicative family has genuine overlap terms and polynomial parameter dependence. These are two distinct deformations, with distinct domains.

PRIME SIEVES / CONSTRUCTIVE WITNESS

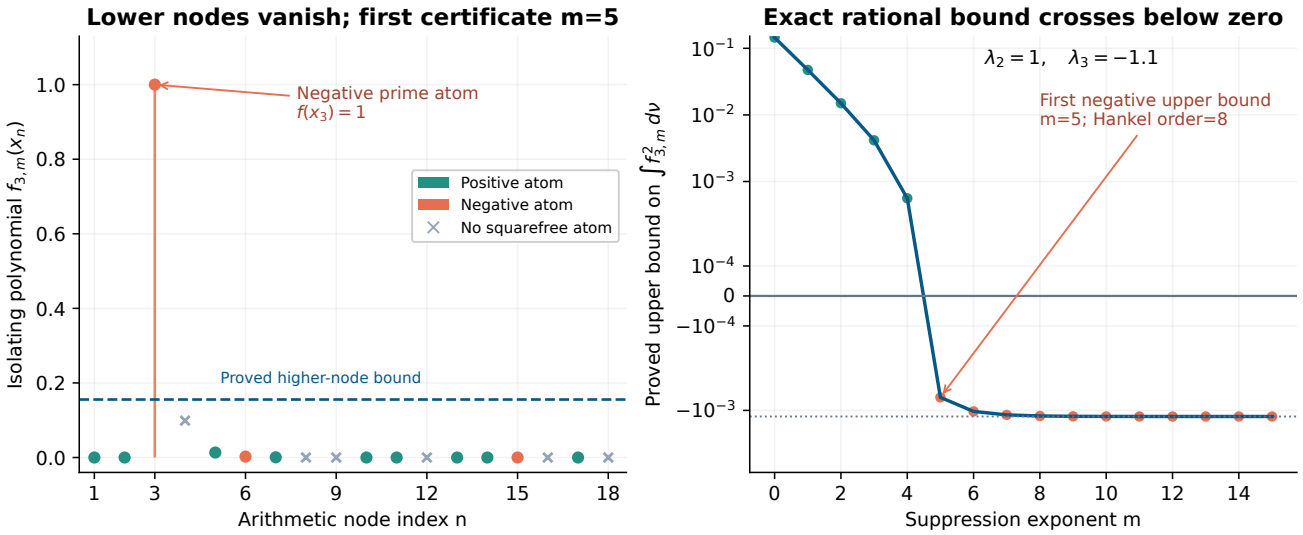
Every negative prime atom has a finite detector

For an integer $a \geq 2$, let

$$L_a(x) = \prod_{n=1}^{a-1} \frac{x - x_n}{x_a - x_n}, \quad C_a = L_a(1), \quad r_a = \frac{1 - x_{a+1}}{1 - x_a} < 1,$$

$$f_{a,m}(x) = L_a(x) \left(\frac{1 - x}{1 - x_a} \right)^m.$$

This polynomial kills every lower node, equals one at x_a, and has absolute value at most $C_a r_a^m$ at every higher node.



A rational detector for $p = 3$, $\lambda_3 = -1.1$, $\lambda_2 = 1$. The first exact bound proving negativity is $m = 5$, so a Hankel matrix of order eight detects the signed defect.

Set $C_F = \prod_{q \in F} \max(1, |1 + \lambda_q|)$. The summable tail gives

$$\int f_{p,m}^2 \, d\nu \leq \frac{1 + \lambda_p}{p^4} + \frac{C_F C_p^2}{3p^3} r_p^{2m}.$$

Hence the quadratic form is negative whenever $pC_F C_p^2 r_p^{2m}/3 < -(1 + \lambda_p)$. For rational strengths this is an exact rational certificate. It needs no numerical eigenvalue calculation. A depth tilt x^k replaces the negative atom contribution by $(1 + \lambda_p) x_p^k / p^4$.

PRIME SIEVES / FIBERS

Real singularities survive multiple deletions

Fix all strengths except λ_p, with the others at least -1. The remaining moment matrix has the form $A_m + (1+\lambda_p)B_m$, where A_m is the p-free part and B_m the p-divisible part. Both are positive definite, even at other deletion endpoints: primes outside F supply the first support, and integers $p\ell$ the second.

Thus every one-variable determinant fiber has exact degree m and all roots real below -1. Its nearest margin satisfies

$$\delta_m = \min_{0\neq f,\deg f<m} \frac{\int f^2\, d\nu_{p\text{-free}}}{\int f^2\, d\nu_{p\text{-divisible}}}, \qquad \delta_m \downarrow 0.$$

The detector yields, for $C_\kappa = \prod_{q\neq p} \max(1, 1+\lambda_q)$,

$$\delta_{p+m} \leq \frac{pC_\kappa C_p^2}{3} r_p^{2m}.$$

The denominator retains the atom p; all lower nodes in the numerator are killed. This extends the prime-two bound to every prime and to simultaneous positive or deletion strengths. It does not assert simple roots or global multivariate real stability.

If at least two positive-domain strengths are nonzero, the complete monic sequence cannot equal a one-prime sequence on the same variable and nodes. Compact moment determinacy equates normalized measures; the fixed mass at zero fixes their scale; atom weights at the individual primes then read off every strength. An arbitrary affine change of variable is outside this separator.

FAMILY 4

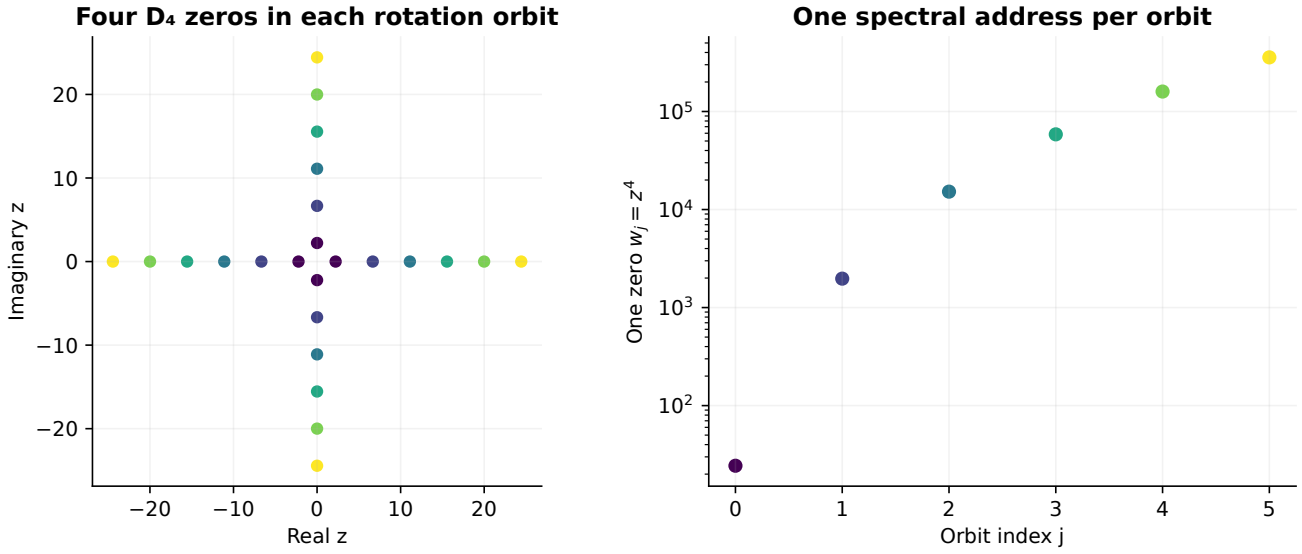
One spectral address for each rotation orbit

For integer $N \geq 2$, define the known Mittag–Leffler seed

$$d_N(w) = \sum_{n\geq 0} \frac{(-1)^n w^n}{(Nn)!} = E_{N,1}(-w), \qquad D_N(z) = d_N(z^N).$$

Let w_j be the nonzero zeros of d_N, counted with multiplicity. Each corresponds to one rotation orbit of N zeros of D_N. Using $\operatorname{Arg} w \in (-\pi, \pi]$, define

$$\boxed{\mathcal{Z}_N(s) = \sum_j w_j^{-s}, \qquad \Re s > 1/N.}$$



For $N = 4$, the four real/imaginary zeros in each orbit share one positive coordinate $w = z^4$. The plot shows the first six orbits. Multiplicity is retained by the definition; simplicity is not assumed in general.

The bound $|d_N(w)| \leq e^{|w|^{1/N}}$ and the finite-order zero-counting theorem give absolute local uniform convergence in the stated half-plane. Since the order is below one and $d_N(0) = 1$, the Hadamard factorization has genus zero:

$$d_N(w) = \prod_j (1 - w/w_j).$$

At positive integers, a log derivative gives

$$-w\frac{d_N'(w)}{d_N(w)} = \sum_{k\geq 1} \mathcal{Z}_N(k) w^k.$$

If $a_n = (-1)^n/(Nn)!$ and $b_k = \mathcal{Z}_N(k)$, then

$$b_k = -k a_k - \sum_{j=1}^{k-1} a_j b_{k-j}.$$

Thus every positive integer value is rational without numerical zero finding.

FAMILY 4 / REDUCTIONS

Classical anchors and the open spectral frontier

The first values are

$$\mathcal{Z}_N(1) = \frac{1}{N!},$$
$$\mathcal{Z}_N(2) = \frac{1}{(N!)^2} - \frac{2}{(2N)!},$$
$$\mathcal{Z}_N(3) = \frac{1}{(N!)^3} - \frac{3}{N!(2N)!} + \frac{3}{(3N)!}.$$

N	$\mathcal{Z}_N(1)$	$\mathcal{Z}_N(2)$	$\mathcal{Z}_N(3)$
2	1/2	1/6	1/15
3	1/6	1/40	53/13440
4	1/24	17/10080	691/9979200
5	1/120	5/72576	997/1743565824

Two exact Hurwitz-zeta reductions

For $N = 2$, $d_2(w) = \cos\sqrt{w}$, whose zeros are $w_j = \pi^2(j + 1/2)^2$, $j \geq 0$. Therefore

$$\mathcal{Z}_2(s) = \pi^{-2s}\zeta(2s, 1/2).$$

For $N = 4$,

$$D_4(z) = \cos(z/\sqrt{2})\cosh(z/\sqrt{2}),$$

and the rotation quotient has one zero $w_j = (\sqrt{2}\pi)^4(j + 1/2)^4$ per orbit. Hence

$$\mathcal{Z}_4(s) = (\sqrt{2}\pi)^{-4s}\zeta(4s, 1/2).$$

These anchors are classical reductions. The general spectral-zeta mechanism also has substantial antecedents, including hypergeometric zero zeta functions.

The continuation gap is closed in 1.0

The original half-plane definition now has a meromorphic extension to the whole complex plane. The next pages prove its sole pole, special values, and regularized product using a branch-safe Mellin representation. These are worked spectral results from established analytic mechanisms. Residue-weighted sums involving derivatives at zeros remain a separate question, especially when multiplicities occur.

ORBIT SPECTRA / 1.0 THEOREM

The continuation gap is closed

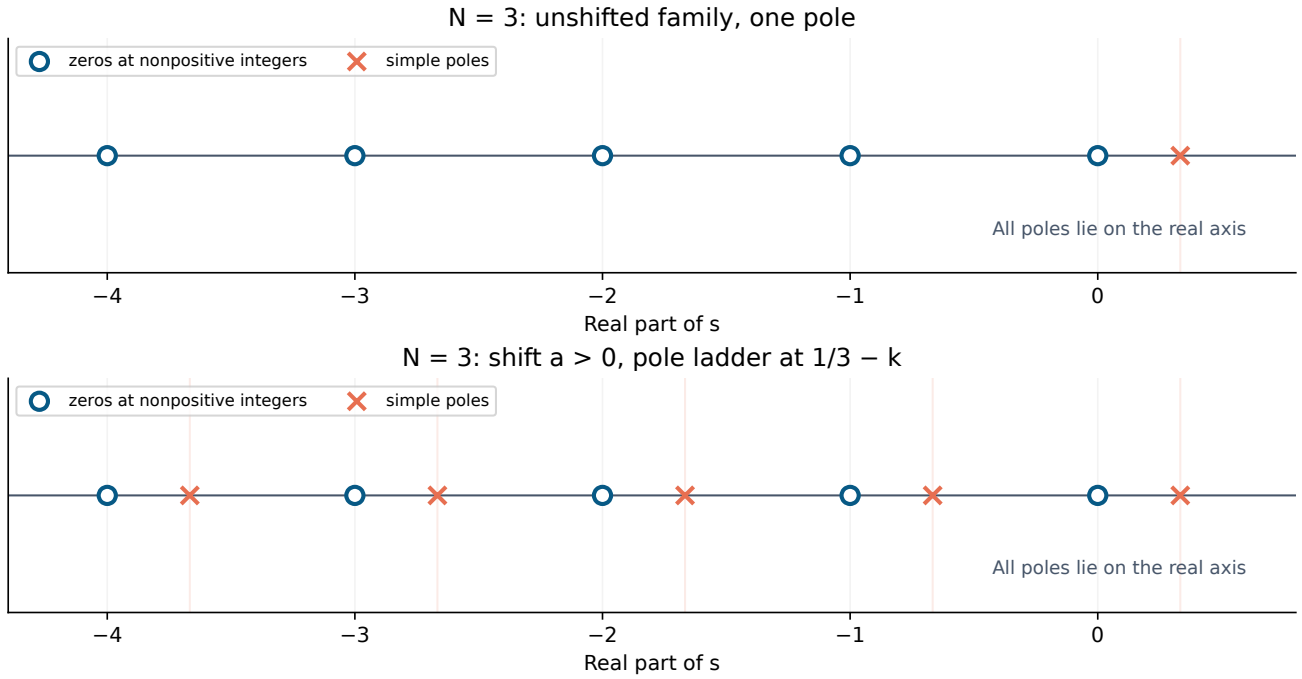
For every integer $N \geq 2$, the orbit zeta has a meromorphic continuation to the whole complex plane with exactly one pole:

$$\boxed{s = 1/N, \qquad \operatorname{Res}_{s=1/N} \mathcal{Z}_N(s) = \frac{\sin(\pi/N)}{N\pi}.}$$

Its exact absolute-convergence abscissa is $1/N$. At nonpositive integers,

$$\boxed{\mathcal{Z}_N(-m) = 0 \quad (m \geq 0), \qquad \mathcal{Z}_N'(0) = -\log N.}$$

The zeta-regularized product of the orbit coordinates is consequently $\exp[-\mathcal{Z}_N'(0)] = N$.



Pole and zero locations follow from the proved continuation. The second panel shows the shifted extension, where a pole ladder appears. These are exact structural locations rather than fitted roots.

The continuation uses classical Mellin regularization. The contribution at this checkpoint is a complete local application with multiplicities, principal branches, and all convergence hypotheses checked.

ORBIT SPECTRA / LARGE RADIUS

A finite exponential sum controls the tail

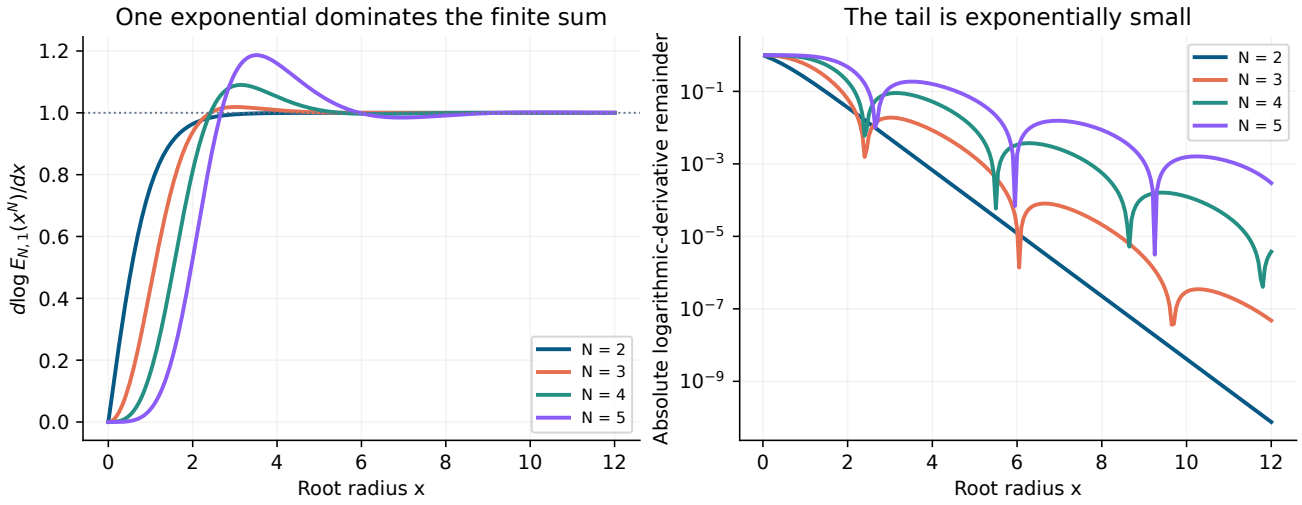
Put $E_N(t) = d_N(-t) = \sum_{n\ge 0} t^n/(Nn)!$ and $\omega = e^{2\pi i/N}$. For $x \ge 0$,

$$E_N(x^N) = \frac{1}{N}\sum_{j=0}^{N-1} e^{\omega^j x}.$$

Root filtering proves this identity term by term: only powers divisible by N remain. The positive real exponential is dominant, so

$$\log E_N(t) = t^{1/N} - \log N + O(e^{-\kappa_N t^{1/N}}), \quad \kappa_N = 1 - \cos(2\pi/N) > 0,$$

with corresponding differentiated estimates on the positive ray.



Computed logarithmic derivatives of the finite exponential sum approach one in the root-radius variable. The decaying remainder supplies the Mellin continuation's large-radius input.

Why complex zero arguments do not break the integral

In a sufficiently narrow sector around negative w, choose the root $x = (-w)^{1/N}$ near the positive real ray. The first exponential dominates the other $N-1$ terms by a uniform real-part gap at large radius, so this sector is eventually zero free. The finitely many remaining zeros avoid the negative real ray because $d_N(-t) > 0$ for $t \ge 0$. Thus all zero arguments have a uniform gap from the branch cut. In particular $|t + w_j| \ge c_N(t + |w_j|)$ for $t \ge 0$.

ORBIT SPECTRA / MELLIN PROOF

The two ends have different jobs

For $1/N < \Re s < 1$, the genus-zero log derivative and the angular gap give absolute interchange in

$$\mathcal{Z}_N(s) = \frac{\sin \pi s}{\pi} \int_0^\infty t^{-s} \frac{d}{dt} \log E_N(t)\, dt.$$

Each term uses the principal-branch beta integral $\int_0^\infty t^{-s}(t+w)^{-1}\, dt = \pi w^{-s}/\sin \pi s$. The denominator estimate bounds the absolute integral by a constant times $|w|^{-\Re s}$, whose sum converges in this strip.

Near zero write $f(t) = (\log E_N)'(t) = \sum_{k\geq 1} a_k t^{k-1}$, where $a_k = (-1)^{k+1}\mathcal{Z}_N(k)$. For any integer $K \geq 1$, subtract K terms on $[0,1]$:

$$\begin{aligned} I_K(s) = &\int_0^1 t^{-s}\left[f(t) - \sum_{k=1}^{K} a_k t^{k-1}\right] dt + \sum_{k=1}^{K} \frac{a_k}{k-s} \\ &+ \int_1^\infty t^{-s}\left[f(t) - \frac{1}{N} t^{1/N-1}\right] dt + \frac{1}{N(s-1/N)}. \end{aligned}$$

The first integral is holomorphic for $\Re s < K+1$; the second is entire because its remainder decays exponentially in $t^{1/N}$. The factor $\sin \pi s$ cancels the small-radius poles at positive integers. Letting K increase covers the plane and leaves only the pole at $1/N$. At nonpositive integers I_K is regular, giving the stated zero values.

At zero the finite parts combine to

$$I_K(0) = \lim_{T\to\infty} [\log E_N(T) - T^{1/N}] = -\log N.$$

Since $(\sin \pi s/\pi)'_{s=0} = 1$, this is $\mathcal{Z}'_N(0)$. The pole also proves the exact abscissa: absolute convergence at any smaller real exponent would give a holomorphic series across $1/N$, a contradiction.

ORBIT SPECTRA / SHIFTS

A Hurwitz-type deformation and its determinant

For real $a \geq 0$, define initially

$$\mathcal{Z}_N(s;a) = \sum_j (w_j + a)^{-s}, \qquad \Re s > 1/N.$$

The rightward shift preserves avoidance of the negative branch cut. Normalize the product with $E_N(a)$ and use $\log E_N(a+t)$ in the Mellin argument. For $a > 0$ there are genuine simple poles at $s = 1/N - k$, $k \geq 0$:

$$\boxed{\operatorname{Res}_{s=1/N-k} \mathcal{Z}_N(s;a) = \frac{\sin(\pi/N)}{N\pi} \frac{(1-1/N)_k}{k!} a^k.}$$

Indeed the large-radius derivative expands as

$$\frac{1}{N}(a+t)^{1/N-1} = \frac{1}{N} \sum_{k\geq 0} \frac{(-1)^k (1-1/N)_k}{k!} a^k t^{1/N-1-k}$$

as an asymptotic series. The alternating sign is canceled by $\sin(\pi(1/N-k)) = (-1)^k \sin(\pi/N)$. At $a = 0$ all secondary residues vanish.

The negative-integer values remain zero, and

$$\mathcal{Z}_N'(0;a) = -\log[NE_N(a)], \qquad \boxed{\prod_j^{\mathrm{reg}} (w_j + a) = NE_{N,1}(a).}$$

This uses the large-radius finite part $-\log N$ and the normalization term $-\log E_N(a)$. It is a zeta-regularized product, not an ordinary convergent product of the coordinates.

For $N = 2$, the result is $2\cosh\sqrt{a}$, a classical half-integer spectral determinant. The four shifted numerical benchmarks reproduce this anchor. Functional equations and derivative-weighted zero sums remain separate questions; the continuation proof does not settle them.

EXACT LABORATORY / COEFFICIENTS

Integer coordinates for the nonlinear family

Normalize the Taylor coefficients by

$$A_n(q,r) = (2n+1)!c_n(q,r), \qquad A_0 = 1.$$

The recurrence becomes

$$\boxed{A_n = \sum_{i=0}^{n-1} \binom{2n}{2i+1} q^{2i+1} r^{2(n-1-i)+1} A_i A_{n-1-i}, \quad n \geq 1.}$$

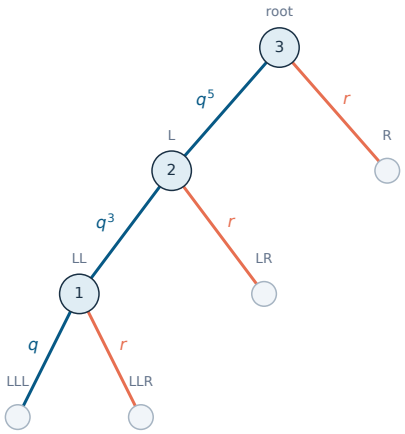
Therefore $A_n \in \mathbb{Z}[q,r]$ has nonnegative coefficients and is symmetric in its two variables. The first coordinates are

$$A_1 = 2qr,$$
$$A_2 = 8q^2r^2(q^2+r^2),$$
$$A_3 = q^3r^3[48(q^2+r^2)(q^4+r^4) + 80q^2r^2].$$

At $(q,r) = (1,1)$ these are the tangent numbers $1, 2, 16, 272, 7936, \ldots$, the known OEIS A000182 anchor. At $(1,2)$ the exact sequence begins $1, 4, 160, 35200, 38041600, \ldots$.

The same three nonlinear nodes have different scale addresses

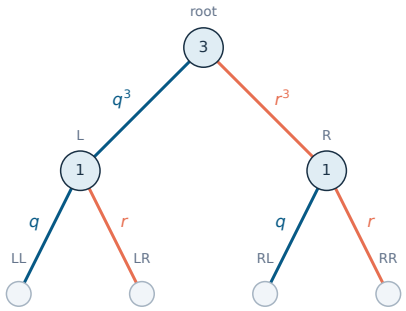
An ordered chain



Tree contribution: $48q^9r^3$

Local integer factors: $6 \times 4 \times 2$

A balanced split



Tree contribution: $80q^5r^5$

Local integer factors: $20 \times 2 \times 2$

At a node with child sizes i, j: edge weights q^{2i+1} and r^{2j+1}; integer factor $\binom{2(i+j+1)}{2i+1}$.

Node labels count internal nodes. Gray leaves carry the base term z. L and R are ordered addresses.

Ordered trees retain left/right path information. The two displayed shapes contribute different monomials to the same factorial-normalized coefficient. Their labels are exact tree data.

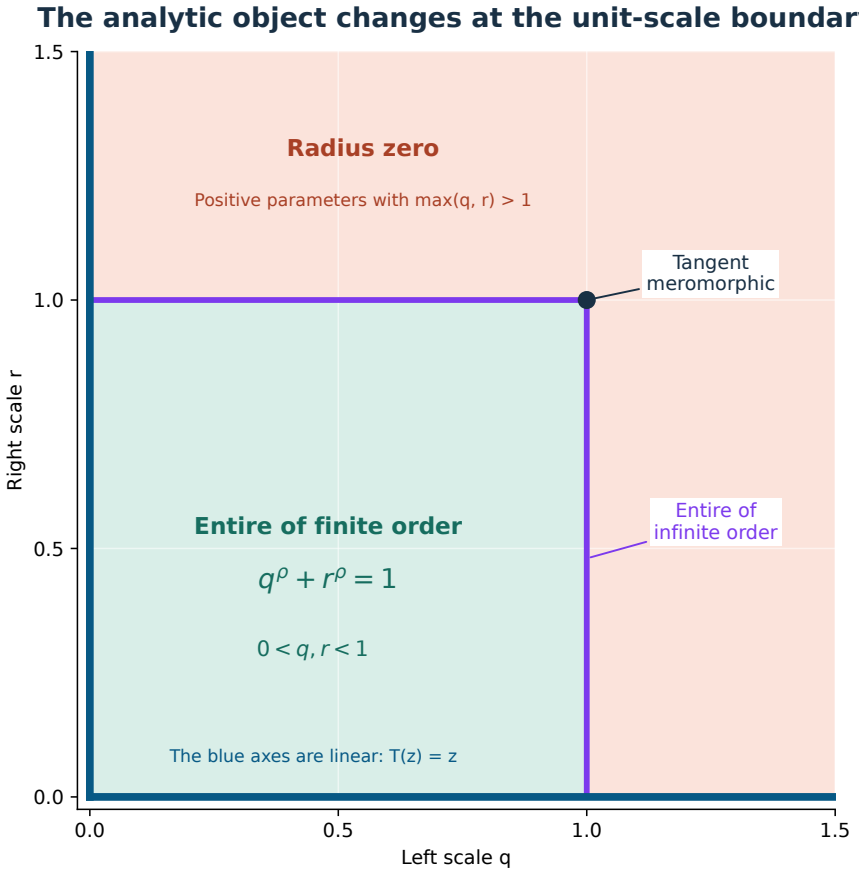
At rational scales, an integer denominator clearing is available. If D clears every contraction denominator and H clears every bank weight,

$$H^n D^{n(n+1)} A_n \in \mathbb{Z}.$$

The maximum root-path exponent sum over a full binary tree is $n(n+1)$. The clearing provides exact sequence exports without implying a new OEIS entry.

EXACT LABORATORY / REGIMES

A formal sequence is not always an analytic germ



The full nonnegative real parameter map separates finite-order entire functions, infinite-order entire edges, tangent, and zero-radius formal series. Zero scales give the linear function.

If $q > 0$, $r > 1$, the recurrence includes its $i = 0$ term and gives

$$c_n \geq \frac{q^n r^{n^2}}{3 \cdot 5 \cdots (2n+1)}.$$

The root test proves zero radius. By symmetry this covers either scale above one, provided the other is positive. The exported positive integer asymmetric rows belong to this formal regime.

If $0 < q < 1$, $r = 1$, the equation is linear in the uncontracted value: $T' = 1 + T(qz)T(z)$. A known coefficient on $|z| < R/q$ gives a linear ODE there, so disk bootstrapping proves entire extension. On the positive ray, $(\log T)' = T(qR) + 1/T(R) \geq qR$, giving an exponential lower bound for T. Substitution back gives, for $R \geq 2/q$,

$$\log T(R) \geq \frac{R}{2} \exp(q^3 R^2/8 - q/2).$$

Thus the entire order is infinite. The corner $(1, 1)$ is tangent with poles. These exact boundaries protect the analytic interpretation of the catalog.

EXACT LABORATORY / PROJECTORS

Polyphase sections preserve the full sequence

For an ordinary generating function $G(z) = \sum_{m \ge 0} a_m z^m$ and integer $b \ge 1$, write

$$G(z) = \sum_{j=0}^{b-1} z^j G_j(z^b), \qquad G_j(t) = \sum_{n \ge 0} a_{bn+j} t^n.$$

The root projector is

$$P_{b,j} G(z) = \frac{1}{b} \sum_{k=0}^{b-1} \omega^{-jk} G(\omega^k z) = z^j G_j(z^b), \qquad \omega = e^{2\pi i/b}.$$

The same projector works on an EGF, but the section retains its original factorials: $\sum_n a_{bn+j} z^{bn+j}/(bn+j)!$. Removing those denominators changes the representation and requires an explicit transport. Root filtering does not do it automatically.

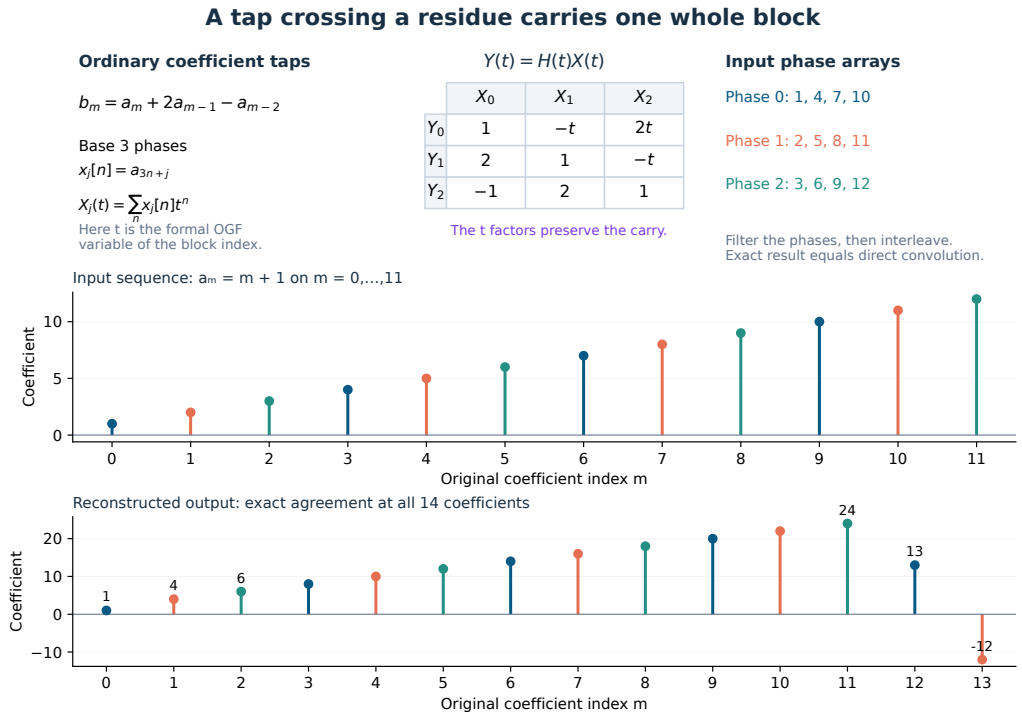
For causal FIR taps h_ℓ, $b_m = \sum_\ell h_\ell a_{m-\ell}$ with zero extension to negative indices, split $A(z)$ and $B(z)$ into b phases. Then $B_i(t) = \sum_j H_{ij}(t) A_j(t)$, where

$$\boxed{H_{ij}(t) = \sum_{\substack{\ell \ge 0 \\ j+\ell \equiv i \pmod b}} h_\ell t^{(j+\ell-i)/b}.}$$

The exponent is the nonnegative carry to the next phase block. This is an exact coefficient law, with no approximation or loss of terms.

EXACT LABORATORY / TAPS

The carry matrix is part of the filter



A three-phase filter with taps $[1, 2, -1]$. The matrix carries terms across block boundaries. Interleaving its output reproduces the direct scalar FIR calculation exactly.

With these taps and $b = 3$,

$$H(t) = \begin{pmatrix} 1 & -t & 2t \\ 2 & 1 & -t \\ -1 & 2 & 1 \end{pmatrix}.$$

The entries arise from the same congruence condition. For instance an input in phase two shifted by one lands in phase zero of the next block, contributing $2t$ to $H_{0,2}$.

Three operations that must be kept distinct

For an EGF $E_a(z) = \sum_m a_m z^m/m!$, differentiation transports future coefficients: $E_a^{(k)} = \sum_m a_{m+k} z^m/m!$. Causal ordinary taps instead act by repeated zero-constant integration:

$$E_b = \sum_\ell h_\ell I^\ell E_a.$$

Multiplication of two EGFs gives Hurwitz convolution, $b_m = \sum_{k=0}^m \binom{m}{k} a_k h_{m-k}$. It is not ordinary sequence FIR convolution. Intrinsic nonlinear feedback taps alter the defining family and are a fourth operation with their own coefficient recurrence.

EXACT LABORATORY / REUSABLE CODE

A catalog with explicit provenance

The standard-library `family_lab.py` exposes exact coefficient polynomials, normalized sequences, positive-bank coefficients, rational denominator clearing, root sections, Hurwitz products, ordinary FIR taps, and polyphase filter matrices. The catalog contains 18 rows through index 12.

Every row records parameters, coefficient normalization, arithmetic regime, and whether its entries are integers or rationals. Eight positive integer asymmetric specializations are labeled zero-radius formal sequences. Entire interior examples and an infinite-order edge are labeled separately.

Example	First normalized coordinates	Interpretation
$(1,1)$	$1,2,16,272,7936$	Known tangent anchor
$(1,2)$	$1,4,160,35200,38041600$	Integer, formal radius zero
$(1/2,1/2)$	$1,1/2,1/4,1/8,1/16$	Scaled sinh, entire order one
$(1/3,2/3)$	$1,4/9,160/729,18560/177147,\ldots$	Asymmetric entire order one

Implementation boundaries are enforced

Phase residues and bases must be integers, and a phase bank cannot be empty. The independent review found that fractional residues and zero phases were previously accepted. Both domain-validation defects were fixed and covered by regression assertions before release.

The 8,038 exact API checks compare separate constructions: Picard integration, sine/cosine tangent division, root filters in a cyclotomic quotient, direct and phase-matrix taps, and coefficient-level antiderivatives. They verify finite implementation ranges. The algebraic laws above have separate proofs in Appendix E.

The catalog is ready for comparison with OEIS, with known anchors identified. It makes no claim that an unmatched prefix establishes a new sequence, and no sequence was submitted externally as part of this release.

RESEARCH JUDGMENT

What 1.0 closes, and what remains open

The 1.0 release is a mathematical and software milestone. It closes selected structural questions and provides a reproducible visual treatment; historical priority remains a separately bounded literature question.

Closed in this release	Still open here
Positive lower and finite upper dilation type	Existence of an optimal type limit and logarithmic modulation
Joint analytic parameter dependence and compact tap bounds	General minimal-bank identifiability
Exact arithmetic parameter recovery up to modulus radical	Stable recovery within bounded prime classes
All-prime signed-atom detectors and positivity domain	Sharp sieve-root asymptotics and multiplicities
Global orbit-zeta continuation and shifted determinants	Functional equations and derivative-weighted zero sums
Integer coefficient and exact polyphase/tap machinery	Historical priority of the candidate extensions

The strongest direct function candidate

The asymmetric dilation extension remains the strongest direct discovery candidate: its parameters are intrinsic, its off-diagonal members cannot be rescaled from the known diagonal, and its nonnegative real scale regime has now been mapped. The arithmetic inverse and signed-resolution results add structural depth. The orbit continuation is a worked application of established spectral methods, with its exact seed and branch conventions verified locally.

A common finite-resolution lesson

A weak tap can change asymptotic order while preserving compact convergence. A large excluded prime can change an exact arithmetic parameter while hiding within a small bounded-window error. A negative atom can evade low-degree positivity and become detectable at a calculable larger degree. These are three different proved mechanisms. Their shared resolution theme does not make them identical constructions.

RELEASE VERIFICATION

Evidence with declared ranges

The independent audit reconstructed the five new proof chapters and checked their final revisions. The release runner executed ten scripts once, with 25 pinned inputs and byte-identical isolated snapshots. All inputs remained unchanged, every declared result was present, and the schema validator passed.

19,871 exact assertion statements + 5,576 numerical assertion statements passed.

The exact total comprises 19,331 principal checks and 540 separately recorded helper checks in the inverse reconstruction. Operational path and process guards are excluded. Counts are assertion statements, not counts of theorems or every scalar operation inside a grouped assertion.

The API contributes 8,038 exact assertions and 18 catalog rows through index 12. Dilation type and continuity add 6,091 exact checks. Arithmetic inverse examples include rank 10,000, negative depth 500, and large excluded primes. The multiple-prime run tests 102 positive parameter/depth cases, 1,395 orthogonality conditions, 381 recurrence cases, and 40 conditional pencils. Spectral continuation tests classical unshifted and shifted anchors.

Recovered failures are retained

The initial 0.1 environment and plotting issues, unsuitable absolute determinant residual, and known one-dilation overlap remain documented. In 1.0, two API validation defects were reproduced and fixed. A draft sequence example had two miscopied integers and was corrected against the exact catalog. Initial figure footer and label overlaps were repaired. A direct publisher-source retrieval failed; an authenticated publisher preview supplied the classical Laplace-uniqueness comparison instead.

Exact analytic tail envelopes are distinguished from floating-point rounding. All decimal arithmetic and root approximations are ordinary high precision, not interval certificates. No analytic Lean certificate is claimed. The universal conclusions rely on the proofs, not on the number of checks.

RELEASE USE

A reproducible research package

The source archive includes the manuscript, all 18 vector figures and PNG previews, figure generators, exact API, JSON/CSV catalog, inherited baseline routes, the five new proof chapters, independent audit, and bounded release evidence with input and output hashes.

The baseline route files preserve the original SF-DISCOVERY-001 claims and open questions as historical evidence. The SF-DISCOVERY-100 chapters supersede those questions where a new proof closes them. The current status is the release matrix above, not an old route paragraph read in isolation.

Reproduce the work

Use the versions listed in `requirements.txt`; execute `checks/run_release.py` in a fresh output directory. The checked instructions give the bounded-run invocation and validator. Figure scripts then rebuild the atlas from formulas and recorded numerical examples. Two `pdflatex` passes compile the source.

The standard-library API is portable. Numerical checks additionally require mpmath. No network access, external submission, or dependency installation is part of the verification run. PDF byte identity can change with timestamps and font environments; mathematical outputs retain their own provenance.

The remaining questions have executable starting points: type-limit analysis for dilation, precision-controlled arithmetic inversion within bounded parameter classes, sharper sieve-root asymptotics, and spectral weighted sums. They remain research questions rather than hidden dependencies of the completed release claims.

SOURCE MAP

Primary references and bounded comparisons

Méndez, Miguel A. *Combinatorial differential operators in: Faà di Bruno formula, enumeration of ballot paths, enriched rooted trees and increasing rooted trees*, *Australasian Journal of Combinatorics* 89 (2024), 350–384. Example 2.4, printed p.356, contains $f' = 1 + f(qz)^2$. https://ajc.maths.uq.edu.au/pdf/89/ajc_v89_p350.pdf

Dascaliuc, Pham, Thomann, Waymire. Primary treatment of the α-Riccati equation $u' = -u + u(\alpha t)^2$, an adjacent feedback model. Its equation is not asserted identical to the asymmetric model here. https://arxiv.org/html/2411.00629v2

Coffey, Mark W. *On the coefficients of the Báez-Duarte criterion for the Riemann hypothesis and their extensions* (2006). Signed Möbius kernels and Newton methods are close antecedents. Their signed weight differs from the positive squarefree weight used here. https://arxiv.org/pdf/math-ph/0608050

NIST DLMF. §16.8, equation 16.8.3, and §16.21, equation 16.21.1, give the differential equations used for the limited D-finite comparison. https://dlmf.nist.gov/16.8 https://dlmf.nist.gov/16.21

NIST DLMF. §18.2 gives general orthogonal-polynomial theory; §10.46, equation 10.46.3, defines the Mittag–Leffler seed. https://dlmf.nist.gov/18.2 https://dlmf.nist.gov/10.46

Uvarov, V. B. *The connection between systems of polynomials that are orthogonal with respect to different distribution functions* (1969). Original metadata was verified; the mass-modification formula is also proved locally in the route file. https://doi.org/10.1016/0041-5553(69)90124-4

Elkies, Noam D. Harvard Math 259 notes on finite-order entire functions and Hadamard products. The genus-zero theorem supports the orbit factorization, with multiplicities retained. https://people.math.harvard.edu/~elkies/M259.02/prod.pdf

Byrnes, Jiu, Moll, Vignat. *Recursion Rules for the Hypergeometric Zeta Function* (2013). Hypergeometric zero spectra and coefficient recursions provide adjacent prior machinery. https://arxiv.org/pdf/1305.1892

Voros, André. *Zeta functions for the Riemann zeros*, Annales de l'Institut Fourier 53(3) (2003), 665–699. Sections 2.1–2.2 describe canonical products, Mellin continuation, and regularization for general sequences. The present orbit proof verifies its own hypotheses. https://www.numdam.org/article/AIF_2003__53_3_665_0.pdf

OEIS. A000182 records the tangent-number anchor. Candidate sequence exports retain parameter, normalization, and regime labels; no OEIS submission or unmatched-prefix novelty claim is made. https://oeis.org/A000182

The literature checks were bounded and performed on 8 October 2026 UTC. Full query lists and source classifications are in the route files. An attempted EMIS retrieval of the original Báez-Duarte paper failed; no theorem here depends on that failed retrieval. Search absence is not a priority proof.

The integer arithmetic seed and its real-depth Gamma–Laguerre comparisons are inherited project material. Their reuse is identified rather than counted again as a discovery.

TECHNICAL APPENDICES

Complete derivations and their dependencies

The visual chapters present the definitions, decisive proof steps, and worked examples. The following appendices retain the complete new derivations, explicit constants, computation contracts, source comparisons, and unresolved questions. Their source files are included unchanged in the archive.

Appendix	Proof chapter	Main dependencies
A	Dilation type and continuity	Positive coefficient recurrence, radius bootstrap, translated-radius comparison
B	Arithmetic inverse and character resolution	Summable positive atoms, squarefree counting, Euler factors, sector partial integration
C	Multiple-prime geometry and detectors	Exact moments, isolated-node polynomials, definite matrix pencils
D	Orbit continuation and shifts	Genus-zero product, branch separation, exponential sums, two-end Mellin subtraction
E	Integer sequences and polyphase laws	Factorial recurrence, ordered trees, root filtering, carry arithmetic

Inherited entire existence, nonrescaling, arithmetic non-D-finiteness, one-prime transport, and base orbit convergence are developed in the visual chapters and in the preserved baseline routes. Those baseline proofs were independently reviewed at SF-DISCOVERY-001. The new audit pins the five complete extensions and exact API used by 1.0.

Classical external leaves are named explicitly. In particular, ordinary orthogonal-polynomial theory, Hadamard factorization, Laplace uniqueness, and spectral Mellin regularization are established machinery. The local proofs verify their application and do not presume historical novelty.

A Positive lower type and quantitative tap continuity

Checkpoint: **SF-DISCOVERY-100**, based on the definitions and entire-extension proof in `routes/asymmetric_dilation.md`. This file strengthens that route's order theorem. It does not alter the classification of the symmetric definition as established, or settle historical priority for the asymmetric extension.

A.1 The exact growth exponent has positive lower type

Let

$$F'(z) = 1 + \sum_{j=1}^{J} h_j F(q_j z) F(r_j z), \qquad F(0) = 0,$$

where $h_j > 0$, $\sum_j h_j = 1$, and $0 < q_j, r_j < 1$. Zero-weight channels may instead be allowed if they are removed from the active bank before defining its growth exponent. Put

$$\rho = \max_{j:h_j>0} \rho_j, \qquad q_j^{\rho_j} + r_j^{\rho_j} = 1.$$

The inherited coefficient recurrence gives nonnegative coefficients, entire extension, and $M(R) = \max_{|z|=R} |F(z)| = F(R) \geq R$ for $R > 0$.

Theorem 1. There are explicit constants $0 < c_- \leq c_+ < \infty$ such that

$$c_- \leq \liminf_{R\to\infty} \frac{\log M(R)}{R^\rho} \leq \limsup_{R\to\infty} \frac{\log M(R)}{R^\rho} \leq c_+.$$

In particular the bank has positive lower type and finite upper type at its exact order, including when $0 < \rho < 1$. Furthermore

$$\lim_{R\to\infty} \frac{\log\log M(R)}{\log R} = \rho.$$

This last radial growth-exponent limit does not assert a type limit or uniform angular growth.

A.1.1 Explicit lower constants and proof

Select any active maximizing channel and abbreviate it to (h, q, r), so $q^\rho + r^\rho = 1$. Define

$$L = \frac{1}{h}, \qquad C = \max\left(\frac{qL}{1-q}, \frac{rL}{1-r}\right), \qquad B = L + \frac{2}{\min(q,r)}, \qquad c_- = \frac{\log 2}{(B+C)^\rho}.$$

We prove the stronger bound

$$\boxed{\log M(R) \geq c_-(R+C)^\rho \quad \text{for every } R \geq 2.}$$

Write $H(R) = \log M(R)$. On the seed interval $[2, B]$, $H(R) \geq \log R \geq \log 2 \geq c_-(R+C)^\rho$. For $R \geq B$, integrate the positive feedback over the fixed-width interval $[R-L, R]$:

$$\begin{aligned} M(R) &\geq h \int_{R-L}^{R} M(qt)M(rt)\,dt \\ &\geq hLM(q(R-L))M(r(R-L)) \\ &= M(q(R-L))M(r(R-L)). \end{aligned}$$

The two child radii are at least two because $R \geq B$. Moreover

$$q(R-L)+C \geq q(R+C), \qquad r(R-L)+C \geq r(R+C),$$

because respectively $(1-q)C \geq qL$ and $(1-r)C \geq rL$. If the comparison bound is known at both child radii, it follows that

$$\begin{aligned} H(R) &\geq H(q(R-L)) + H(r(R-L)) \\ &\geq c_-\{[q(R-L)+C]^\rho + [r(R-L)+C]^\rho\} \\ &\geq c_-(q^\rho + r^\rho)(R+C)^\rho = c_-(R+C)^\rho. \end{aligned}$$

For a rigorous propagation, put $Q_* = \max(q,r) < 1$ and $d = (1-Q_*)B + Q_*L > 0$. Every child of every $R \geq B$ is at most $R-d$. Induction on the intervals $[B+kd, B+(k+1)d]$, beginning with the already controlled seed, proves the comparison for all larger radii. Continuity deals with common endpoints. All power comparisons use only monotonicity of x^ρ for positive x; no convexity assumption is made, so the argument also covers $\rho < 1$.

A.1.2 Explicit finite upper constants

For all channels set

$$Q = \max_j \max(q_j, r_j), \quad m = \min_j \min(q_j, r_j), \quad \Delta = \log(1/m), \quad B_+ = \max_j \log \frac{1}{q_j r_j} + \log 2.$$

Choose an integer $N \geq 0$ with $Q^{-N} \geq 3/m$, and put

$$V_N = (2^{N+1}-1)\log 2 + (2^{N+1}-N-2)\log(1/Q),$$

$$c_+ = e^{-\rho}(V_N + 1 + \Delta + B_+).$$

These constants are intentionally conservative, and require no numerical evaluation of an unknown solution. The tangent majorant gives $M(1) < 2$. At $R_k = Q^{-k}$, the integral equation gives

$$M(R_k) \leq R_k[1 + M(R_{k-1})^2].$$

Define $U_0 = 2$, $U_k = Q^{-k}(1+U_{k-1}^2)$. Then $M(R_k) \leq U_k$, $U_k \geq 1$, and $\log U_k \leq k\log(1/Q) + \log 2 + 2\log U_{k-1}$. Summing this elementary recurrence gives $\log U_N \leq V_N$. Since $e < 3$, the logarithmic-radius seed $1 \leq t \leq 1+\Delta$ has $e^t \leq e/m < 3/m \leq R_N$, and therefore $\log M(e^t) \leq V_N$.

Let $a_j = \log(1/q_j)$, $b_j = \log(1/r_j)$, and $G(t) = \log M(e^t) + t + B_+$. For $t \geq 1+\Delta$, all child moduli are at least one, so the positive integral and weight normalization give

$$\log M(e^t) \le t + \log 2 + \max_j \{\log M(e^{t-a_j}) + \log M(e^{t-b_j})\}.$$

The definition of B_+ implies

$$G(t) \le \max_j \{G(t-a_j) + G(t-b_j)\}.$$

On the seed, $G(t) \le V_N + 1 + \Delta + B_+ \le c_+ e^{\rho t}$. Each channel satisfies $q_j^\rho + r_j^\rho \le 1$. Induction in intervals of length $\min_j \min(a_j, b_j) > 0$ therefore propagates the seed comparison, giving the explicit upper bound

$$\boxed{\log M(R) \le c_+ R^\rho - \log R - B_+ \quad (R \ge e).}$$

Dividing the two bounds by R^ρ, then taking logarithms of their positive logarithmic-growth bounds, proves all assertions of Theorem 1. The theorem leaves open whether $\log M(R)/R^\rho$ has a limit. Lattice resonances or logarithmic oscillations are possible research directions, not results here.

A.2 Joint holomorphic dependence on parameters

Fix the number J of channels. Allow complex weights h_j, without normalization, and complex contractions satisfying $|q_j|, |r_j| < 1$. The same formal recurrence defines coefficients

$$c_n(h, q, r) \in \mathbb{Q}[h_1, \dots, h_J, q_1, r_1, \dots, q_J, r_J].$$

Each is homogeneous of degree n in the weights. If a parameter set has $|q_j|, |r_j| \le Q < 1$ and $\sum_j |h_j| \le W$, with $W > 0$, coefficient induction gives

$$|c_n(h, q, r)| \le W^n d_n(Q),$$

where $f_Q(z) = \sum d_n(Q) z^{2n+1}$ is the established diagonal function. Consequently, on every disk $|z| \le R$,

$$\sum_{n \ge 0} |c_n(h, q, r)| R^{2n+1} \le W^{-1/2} f_Q(\sqrt{W} R) < \infty.$$

Every compact subset of the complex parameter domain has such Q, W. The series therefore converges normally on compact sets of the joint domain, and its sum is jointly holomorphic in the weights, contractions, and z. This proves parameter continuity at zero weights, although the growth order can jump there. Complex or signed parameters do not inherit the positive-bank growth theorem merely from holomorphic dependence.

A.3 A quantitative compact bound for changing tap weights

Keep fixed real contractions $0 < q_j, r_j \le Q < 1$, and let h, g be nonnegative normalized weight vectors. Zero weights are permitted. Let $d = \|h - g\|_1$. Then for every $R \ge 0$,

$$\boxed{\sup_{|z| \le R} |F_h(z) - F_g(z)| \le \frac{d}{4} \{R f_Q'(R) - f_Q(R)\}.}$$

Proof. Regard c_n as a polynomial in the real weight variables with the contractions fixed. Its coefficients are nonnegative. Induction in the recurrence gives the coefficientwise polynomial bound

$$0 \preceq c_n(h) \preceq d_n(Q)\Big(\sum_j h_j\Big)^n.$$

Here $\preceq$ compares every monomial coefficient. Differentiation preserves this comparison, so on the normalized nonnegative simplex

$$0 \le \partial_{h_j} c_n \le n d_n(Q).$$

Put $v = h - g$. Its positive and negative parts both have sum $d/2$, since $\sum_j v_j = 0$. At every point of the segment between g and h, all partial derivatives lie in the interval $[0, nd_n(Q)]$. Hence

$$|v \cdot \nabla c_n| \le \frac{d}{2} n d_n(Q), \qquad |c_n(h) - c_n(g)| \le \frac{d}{2} n d_n(Q).$$

Multiply by R^{2n+1} and sum. Entire convergence justifies the sum, and

$$R f_Q'(R) - f_Q(R) = 2 \sum_{n \ge 0} n d_n(Q) R^{2n+1},$$

proving the claim. The same argument applies to the closed simplex and needs no positive minimum weight.

A.3.1 A vanishing tap and discontinuous order

Take the two diagonal channels $(q, r) = (1/4, 1/4)$ and $(3/4, 3/4)$, with weights $1 - \varepsilon$ and ε, respectively. For $0 \le \varepsilon \le 1$, write the resulting function as F_ε. The quantitative bound gives

$$\sup_{|z| \le R} |F_\varepsilon(z) - f_{1/4}(z)| \le \frac{\varepsilon}{2} \{R f_{3/4}'(R) - f_{3/4}(R)\}.$$

Thus convergence to $f_{1/4}$ is uniform on every fixed disk, at an explicit linear rate in ε. Yet

$$\rho(F_0) = \frac{1}{2}, \qquad \rho(F_\varepsilon) = \frac{\log 2}{\log(4/3)} \quad (0 < \varepsilon \le 1).$$

For the faster active channel the lower constants become $L = 1/\varepsilon$, $C = 3/\varepsilon$, $B = 1/\varepsilon + 8/3$. Its certified lower-type constant is

$$c_-(\varepsilon) = \frac{\log 2}{(4/\varepsilon + 8/3)^{\log 2 / \log(4/3)}} > 0.$$

It tends to zero as the faster tap vanishes. These are proved growth and compact-convergence statements. They do not determine an actual crossover radius or leading asymptotic constant.

A.4 Computational contract and limitations

`checks/dilation_type.py` uses exact `fractions.Fraction` arithmetic for its finite assertions. It checks the algebraic fixed-width and interval-propagation conditions for rational scales and weights, the positive homogeneous weight-polynomial majorant, derivative bounds on a normalized simplex, the improved total-variation coefficient bound, and a distinct ordered-tree evaluation against the recurrence. Scalar examples and the explicit real type constants are displayed at high precision separately from exact assertions.

The finite checks do not prove a universal analytic theorem, normal convergence, or historical novelty. Those depend on the proofs above. No numerical type-limit fit is presented as evidence that a limit exists.

The earlier moving-width lower argument used a fixed $\alpha < 1$; it proved orders below ρ and then recovered the order by $\alpha \uparrow 1$, but did not establish positive lower type at ρ. This is a limitation of that argument, not a counterexample. The fixed-width comparison closes precisely that missing implication. Any execution failure of the new finite checks is recorded with its cause and retained run metadata rather than erased.

B Arithmetic depth: an exact inverse, its conditioning, and character resolution

Checkpoint: SF-DISCOVERY-100. Attribution: Komorebi P.

The entire squarefree depth function determines its arithmetic parameters exactly, with one precise equivalence: powers of the primes in the modulus are invisible. This is a local mathematical theorem. It does not assert historical priority for Laplace inversion or for this particular arithmetic application.

B.1 Claim card and conventions

For real $a > 1/2$, real $\beta > 0$, and integer $Q \geq 1$, write

$$A_{a,\beta,Q}(s) = \sum_{\substack{n\geq 2\\ \mu(n)^2=1\\ (n,Q)=1}} n^{-2a}\exp(-\lambda_n s), \qquad \lambda_n = -2\log(1-n^{-\beta}).$$

The logarithm is real. The omitted $n = 1$ atom is not restored in any inverse formula. Let $n_1 < n_2 < \cdots$ be the support integers, and abbreviate $\lambda_j = \lambda_{n_j}$, $w_j = n_j^{-2a}$. Then $\lambda_1 > \lambda_2 > \cdots > 0$, $\lambda_j \to 0$, and $\sum_j w_j = M = Z_Q(2a) - 1 < \infty$, where

$$Z_Q(v) = \frac{\zeta(v)}{\zeta(2v)}\prod_{p\mid Q}(1+p^{-v})^{-1}, \qquad v > 1.$$

Here a germ means the complete exact analytic germ, equivalently all exact derivatives at a point, rather than a finite Taylor polynomial or noisy measurements. The base route proves that this germ has a unique entire continuation, with $|A(s)| \leq M\exp(\lambda_1 \max\{-\Re s, 0\})$.

B.2 Exact inverse theorem

Theorem AI-1. Two admissible parameter triples define the same analytic germ if and only if

$$a = a', \qquad \beta = \beta', \qquad \operatorname{rad}(Q) = \operatorname{rad}(Q').$$

The function itself therefore encodes the weight exponent, the depth exponent, and the finite excluded prime set. It does not encode valuations $v_p(Q)$.

B.2.1 Step 1: recover the ordered spectral atoms

Starting with $R_1 = A$, define successively

$$\lambda_j = \lim_{t\to\infty}\frac{\log R_j(-t)}{t}, \qquad w_j = \lim_{t\to\infty} e^{-\lambda_j t}R_j(-t), \qquad R_{j+1}(s) = R_j(s) - w_j e^{-\lambda_j s}.$$

Every limit exists and recovers exactly the j-th support frequency and weight. Indeed, if $M_{j+1} = \sum_{k>j} w_k$, then for every $t \geq 0$,

$$R_j(-t) = w_j e^{\lambda_j t}(1 + r_j(t)), \qquad 0 \leq r_j(t) \leq \frac{M_{j+1}}{w_j}e^{-(\lambda_j - \lambda_{j+1})t}.$$

The bound follows by replacing every tail frequency by its largest value λ_{j+1}. Taking logarithms and dividing by t proves the first limit. Multiplying by $e^{-\lambda_j t}$ proves the second. Removing this exact entire atom leaves the same hypotheses for the next step. Thus the complete ordered atom list is intrinsic to the entire function.

No approximation to a large negative-depth value is used in this uniqueness proof. It is an exact limiting argument.

B.2.2 Step 2: recover β without knowing the integer support

Put

$$\delta_Q = \frac{1}{\zeta(2)} \prod_{p \mid Q} \frac{p}{p+1} > 0.$$

The base route establishes the elementary counting estimate

$$N_Q(x) := \#\{2 \le n \le x : \mu(n)^2 = 1, (n, Q) = 1\} = \delta_Q x + O_Q(\sqrt{x}).$$

Since $N_Q(n_j) = j$, it follows that

$$n_j = \frac{j}{\delta_Q} \left(1 + O_Q(j^{-1/2})\right), \quad \lambda_j = 2\delta_Q^{\beta} j^{-\beta} \left(1 + O_{Q,\beta}(j^{-1/2} + j^{-\beta})\right).$$

For completeness, the first estimate follows initially from $j = \delta_Q n_j + O_Q(\sqrt{n_j})$, which gives $n_j \asymp_Q j$; substituting this back improves the error to $O_Q(\sqrt{j})$. The second uses $-\log(1-u) = u(1+O(u))$ as $u \downarrow 0$.

Consequently the already recovered frequencies determine

$$\boxed{\beta = -\lim_{j \to \infty} \frac{\log \lambda_j}{\log j}.}$$

A rank ratio avoids the slowly decaying constant divided by $\log j$: for every fixed real $c > 1$,

$$-\frac{\log(\lambda_{\lfloor cj \rfloor} / \lambda_j)}{\log c} = \beta + O_{Q,\beta,c}(j^{-1/2} + j^{-\beta}).$$

The floor contributes an additional $O(j^{-1})$, which is absorbed by the displayed error. These are fixed-parameter asymptotics. They do not give a constant uniform in arbitrary moduli.

B.2.3 Step 3: recover the support integers and the weight exponent

Once β is known, every spectral atom gives its exact integer label:

$$\boxed{n_j = (1 - e^{-\lambda_j/2})^{-1/\beta}, \qquad a = -\frac{\log w_j}{2 \log n_j}.}$$

The recovered real number in the first formula is an integer by the family definition; rounding is not part of the exact theorem. Since $n_j \geq 2$, the denominator in the second formula is nonzero. Every recovered atom yields the same value of a.

B.2.4 Step 4: recover the excluded primes

A prime p lies in the recovered integer support exactly when $p \nmid Q$: primes are squarefree, and no other restriction is imposed. Hence

$$\boxed{\operatorname{rad}(Q) = \prod_{\substack{p \text{ prime} \\ p \notin \{n_j : j \geq 1\}}} p.}$$

The product is finite because the parameters are assumed to come from this family. This formula is an exact reconstruction from infinite data, not an algorithm that can certify completion of the prime list after finitely many observations. Conversely, equal a, β and equal radicals produce exactly the same terms. This proves both directions of AI-1.

B.2.5 A constructive counting envelope

The rank asymptotic can be made effective without an RH input. Write $\omega(Q) = \#\{p : p \mid Q\}$. For real $x \geq 4$, inclusion and exclusion gives

$$|N_Q(x) - \delta_Q x| \leq C_Q \sqrt{x} + 1, \qquad C_Q = 2^{\omega(Q)} + 2\frac{\varphi(Q)}{Q}.$$

To verify the constants, expand $\mu(n)^2 = \sum_{d^2 \mid n} \mu(d)$. For $(d, Q) = 1$, the number of $m \leq x/d^2$ coprime to Q differs from $(\varphi(Q)/Q)x/d^2$ by at most $2^{\omega(Q)}$. Summing over at most $\sqrt{x}$ values of d gives the first part of C_Q. The omitted main-term tail is at most $(\varphi(Q)/Q)x/\lfloor\sqrt{x}\rfloor \leq 2(\varphi(Q)/Q)\sqrt{x}$. Removing $n = 1$ adds one. The main-term density is $(\varphi(Q)/Q)\sum_{(d,Q)=1} \mu(d)d^{-2} = \delta_Q$ by its absolutely convergent Euler product.

For $j \geq 3$, therefore,

$$\left(\frac{\sqrt{C_Q^2 + 4\delta_Q(j-1)} - C_Q}{2\delta_Q}\right)^2 \leq n_j \leq \left(\frac{\sqrt{C_Q^2 + 4\delta_Q(j+1)} + C_Q}{2\delta_Q}\right)^2.$$

The lower and upper bounds are the positive solutions of the corresponding quadratic inequalities in $\sqrt{n_j}$. At $j \geq 3$, all possible supports have $n_j \geq 5$, so the $x \geq 4$ counting envelope applies.

B.3 What finite negative-depth measurements can recover

Theorem AI-2, gap bound. Suppose the first $j-1$ atoms have already been removed exactly, put $\Delta_j = \lambda_j - \lambda_{j+1} > 0$, and fix $h > 0$. Define

$$\widehat{\lambda}_j(t, h) = \frac{\log R_j(-t-h) - \log R_j(-t)}{h}.$$

Then for every $t \geq 0$,

$$0 \leq \lambda_j - \widehat{\lambda}_j(t,h) \leq \frac{1}{h}\log\left(1 + \frac{M_{j+1}}{w_j}e^{-\Delta_j t}\right),$$

and

$$0 \leq e^{-\lambda_j t}R_j(-t) - w_j \leq M_{j+1}e^{-\Delta_j t}.$$

Proof. The preceding positive remainder formula gives $\widehat{\lambda}_j = \lambda_j + h^{-1}[\log(1 + r_j(t+h)) - \log(1 + r_j(t))]$. Each tail factor decreases, so $0 \leq r_j(t+h) \leq r_j(t)$. The two inequalities now follow from the bound on $r_j(t)$. For the amplitude, multiply the positive tail bound by $e^{-\lambda_j t}$.

In particular, to guarantee frequency error at most $\varepsilon > 0$ from exact residual values, it suffices to take

$$t \geq \max\left\{0, \frac{1}{\Delta_j}\log\frac{M_{j+1}}{w_j(e^{h\varepsilon} - 1)}\right\}.$$

The parameters in this certificate are properties of the true tail. If they are only estimated from data, this is a conditional recovery bound rather than an independent numerical certificate.

B.3.1 Positive tail envelopes for reproducible examples

For integer $L \geq n_j$, a direct sum through $n \leq L$ at depth $-t$, $t \geq 0$, omits at most

$$E_L(t) = e^{t\lambda_{L+1}}\frac{L^{1-2a}}{2a-1}.$$

Indeed all omitted frequencies are at most λ_{L+1}, and $\sum_{n>L} n^{-2a} \leq \int_L^\infty x^{-2a}\,dx$. Ignoring squarefreeness and coprimality makes this a valid upper bound. If r_0, r_1 are the two positive truncated residual sums at $t, t+h$, the true secant belongs to

$$\left[\frac{\log r_1 - \log(r_0 + E_L(t))}{h}, \frac{\log(r_1 + E_L(t+h)) - \log r_0}{h}\right].$$

This interval treats truncation exactly at the mathematical level. Its displayed floating-point endpoints are not directed-rounding certificates.

B.3.2 Cancellation is a real limitation

Evaluating a residual by subtracting previous atoms is different from evaluating its known positive series. If an observation of $A(-t)$ has relative error at most η and the previous atoms are exact, then the residual error can have relative size

$$\eta\,\kappa_j(t), \qquad \kappa_j(t) = \frac{A(-t)}{R_j(-t)} \sim \frac{w_1}{w_j}e^{(\lambda_1 - \lambda_j)t}.$$

Thus the large negative depth that separates nearby atoms also worsens cancellation for later atoms. Uncertain previously subtracted frequencies introduce further error. The computation evaluates residuals as positive sums and reports $\log_{10} \kappa_j$ so this conditioning cost remains visible.

B.4 Exact uniqueness does not imply uniform prime recovery

Theorem AI-3, bounded-window instability. Let p be a prime not dividing Q, and let $R \geq 0$. For every integer $k \geq 0$,

$$\boxed{\sup_{|s|\leq R} |A^{(k)}_{a,\beta,Q}(s) - A^{(k)}_{a,\beta,Qp}(s)| \leq \lambda_p^k e^{R\lambda_p} \frac{Z_Q(2a)}{p^{2a}+1}.}$$

Consequently adding larger and larger excluded primes gives convergence in every derivative on every bounded complex-depth window, even though the recovered radical changes each time.

Proof. The difference consists exactly of the squarefree integers divisible by p and coprime to Q, all of which satisfy $n \geq p$. Their frequencies are at most λ_p. The Euler factor at p shows that their total weight is

$$\sum_{\substack{n\geq 1 \\ \mu(n)^2=1,(n,Q)=1 \\ p|n}} n^{-2a} = \frac{Z_Q(2a)}{p^{2a}+1}.$$

There is no $n = 1$ term in this difference. Bound the k-th derivative of every exponential by $\lambda_p^k e^{R\lambda_p}$ and sum. Since $\lambda_p \to 0$ and $p^{-2a} \to 0$, the bound tends to zero. This proves the derivative convergence and the claimed instability.

More explicitly, for any bounded window and positive measurement tolerance, two distinct radicals in this family can produce functions closer than that tolerance throughout the window. This defeats a recovery guarantee uniform over unbounded excluded primes. It does not refute recovery under an a priori modulus bound or from complete exact analytic data.

For $a = \beta = 2, Q = 1$, the mass removed by p is exactly $105/[\pi^4(p^4+1)]$. The examples $p = 101, 1009, 10007$ quantify how quickly large-prime distinctions disappear on $|s| \leq 5$.

B.5 The function as an arithmetic Hilbert feature space

This section spells out the classical feature-space construction for this arithmetic spectrum. It supplies a precise space in which the complex-depth packets live.

For $c = (c_j) \in \ell^2$, define

$$(Uc)(s) = \sum_{j\geq 1} c_j \sqrt{w_j}\, e^{-\lambda_j s}.$$

Cauchy and Schwarz give uniform convergence on bounded complex sets and

$$|(Uc)(s)| \leq \|c\|_{\ell^2} \sqrt{A(2\Re s)}.$$

The map is injective: $\sum |c_j|\sqrt{w_j} \le \|c\|\sqrt{M} < \infty$, so the same spectral peeling argument, now for complex coefficients, forces the first coefficient and then every coefficient to vanish whenever $Uc = 0$. Positivity was needed to recover unknown frequencies in AI-1; injectivity here only needs absolute summability and the known ordered support. Specifically, after dividing by the largest remaining exponential, the tail is bounded by $e^{-\Delta_j t}\sum_{k>j}|c_k|\sqrt{w_k}$, which tends to zero.

The image $\mathcal{H}_{a,\beta,Q} = U\ell^2$, with $\|Uc\|_{\mathcal{H}} = \|c\|_{\ell^2}$, is therefore a complete Hilbert space of entire functions. Its reproducing kernel is

$$K(s,t) = A(s+\bar{t}).$$

Using an inner product linear in the first argument, the coefficient vector of $K(\cdot,t)$ is $(\sqrt{w_j}e^{-\lambda_j\bar{t}})_j$, so $\langle Uc, K(\cdot,t)\rangle = (Uc)(t)$. This also verifies the sign and conjugation convention directly. Arbitrarily large finite Gram matrices at distinct complex depths are strictly positive, as proved in the base route. Thus the feature space has infinite rank, not a finite-dimensional Prony model.

The monomial moment matrices also have infinite rank: $m_k = (-1)^k A^{(k)}(0) = \sum_j w_j\lambda_j^k$, and for every nonzero polynomial P, $\sum_j w_j|P(\lambda_j)|^2 > 0$, because a nonzero polynomial cannot vanish at infinitely many distinct frequencies. All finite derivative Hankel matrices are strictly positive definite.

B.6 Complete fixed-modulus character resolution

Let χ, ψ be Dirichlet characters modulo the same fixed integer Q, extended by zero outside the units, and put $\eta = \chi\overline{\psi}$. The base extension is

$$A_{\chi,\psi}(s) = \sum_{n\ge 2}\mu(n)^2\eta(n)n^{-2a}(1-n^{-\beta})^{2s}.$$

The endpoint subtraction in its Newton expansion is still exactly one:

$$A_{\chi,\psi}(s) = \sum_{j\ge 0}(-1)^j\binom{2s}{j}\left[\frac{L(2a+j\beta,\eta)}{L(4a+2j\beta,\eta^2)} - 1\right].$$

Every character takes the value one at $n = 1$; at primes dividing Q, it is zero. The Euler product identity is valid even when η is imprimitive, because the actual Dirichlet character and its actual modulus are retained. All arguments of L lie in $\Re v > 1$, where its Euler product converges absolutely and is nonzero. No unproved zero-free-region hypothesis is used.

Theorem AI-4, character sector decay. Fix $a > 1/2, \beta > 0, Q$, and a closed sector $|\arg s| \le \pi/2 - \varepsilon$, $0 < \varepsilon < \pi/2$. If $\chi \ne \psi$, then uniformly in that sector,

$$A_{\chi,\psi}(s) = O_{a,\beta,Q,\varepsilon}(|s|^{-(2a-1/2)/\beta}), \qquad \frac{A_{\chi,\psi}(s)}{A_{a,\beta,Q}(s)} = O_{a,\beta,Q,\varepsilon}(|s|^{-1/(2\beta)}).$$

These statements are uniform over the finite character set of this fixed modulus. They are not uniform as Q grows.

Proof. Distinct characters give a nonprincipal η. Its sum over a complete period is zero, so $|\sum_{m\le y}\eta(m)| \le Q$. Hence

$$\sum_{n\le x}\mu(n)^2\eta(n)=\sum_{d\le\sqrt{x}}\mu(d)\eta(d)^2\sum_{m\le x/d^2}\eta(m)=O_Q(\sqrt{x}).$$

Use partial integration against $g_s(x)=x^{-2a}(1-x^{-\beta})^{2s}$, whose endpoint values vanish when $\Re s>0$. The counting bound reduces the magnitude to a constant times $\int_1^\infty\sqrt{x}\,|g_s'(x)|\,dx$. Writing $u=\Re s$ and $\gamma=(2a-1/2)/\beta>0$, direct substitution $y=x^{-\beta}$ bounds its two derivative terms by

$$\frac{2a}{\beta}B(\gamma,2u+1)+2|s|B(\gamma+1,2u),$$

which is $O_{a,\beta,\varepsilon}(|s|^{-\gamma})$ because $u\ge|s|\sin\varepsilon$. The principal diagonal has the nonzero leading term $\delta_Q\Gamma(\alpha)\beta^{-1}(2s)^{-\alpha}$, $\alpha=(2a-1)/\beta$, by the base sector theorem. Its modulus is bounded below by a positive fixed-sector multiple of $|s|^{-\alpha}$ for sufficiently large $|s|$. Division gives the second bound, since $\gamma-\alpha=1/(2\beta)$.

At real positive depth, the matrix $(A_{\chi,\psi}(s))$ on any d distinct characters is positive semidefinite and has identical diagonal $A_{a,\beta,Q}(s)$. Its normalized matrix tends to I_d, with operator norm error $O_{a,\beta,Q,d}(s^{-1/(2\beta)})$. Indeed, the off-diagonal entry bound controls the maximum row and column sums, and thus the operator norm. This gives quantitative separation of a fixed character bank. It is consistent with a complex or signed off-diagonal entry and makes no complete-monotonicity assertion for that entry.

B.7 Classical comparison and limits of novelty

The general fact that a positive measure is determined by its Laplace transform is classical. The primary publisher-distributed preview of René L. Schilling, Renming Song, and Zoran Vondraček, *Bernstein Functions: Theory and Applications*, first edition, de Gruyter, 2010, ISBN 978-3-11-021530-4 / electronic ISBN 978-3-11-021531-1, was checked on 2026-10-08 UTC:

`https://api.pageplace.de/preview/DT0400.9783110215311_A15706968/preview-9783110215311_A15706968.pdf`

Proposition 1.2, printed page 2 / PDF page 14 under one-based indexing, states uniqueness of a measure on $[0,\infty)$ from its Laplace transform and proves recovery from all derivatives. Theorem 1.4, printed page 3 / PDF page 15, is the Bernstein representation theorem. Translation here: the source's measure is $\sum_j w_j\delta_{\lambda_j}$, its transform variable is the present real depth $s>0$, and its finiteness condition holds because $\sum_j w_j<\infty$. Classification: general uniqueness is **known after translation of notation**. The exact extraction of $a,\beta,\mathrm{rad}(Q)$, the rank asymptotic, and the displayed large-prime instability are proved locally rather than claimed as results of that book.

The Hilbert feature-space construction and finite Prony mechanism are also classical constructions. They are explicit realizations for this arithmetic spectrum, with no priority claim. The parameter-specific inverse theorem and character extension are not asserted historically new. The limited searches do not settle whether these results have appeared together for this exact positive squarefree family.

Bounded discovery queries were `squarefree Laplace transform inverse identification parameters spectral measure arithmetic` and `Bernstein functions theorem 1.4 completely monotone uniquely determines measure Schilling Song Vondracek pdf`. The direct de Gruyter chapter URL returned an internal retrieval error. The publisher-distributed preview supplied the needed theorem and full proof instead. No conclusion depends on the failed URL, and no locally cached publisher PDF is included in the release. The elementary inverse and character proofs do not require this external source as an

unchecked leaf.

B.8 Computation contract and handoff

`checks/arithmetic_inverse.py` is deterministic. Its exact checks use rational finite atomic surrogates with $2a = 4$, integer $\beta = 1, 2, 3$, $Q = 1, 2, 6, 30$, and one through five atoms. They independently construct the annihilating polynomial from moments and from its known rational nodes, verify the positive Vandermonde formula for Hankel determinants, reconstruct weights, recover integer labels at known β, and test further exact recurrence moments. These surrogates do not turn the infinite family's Hankel matrices into finite-rank matrices.

The high-precision examples use 180 decimal digits. They show two-rank β estimates at ranks 100, 1000, and 10000, conditional negative-depth peeling for the first three atoms at $t = 20, 100, 500$, and large-prime bounded-window envelopes. Positive direct sums use $n \leq 5000$, with the analytic omitted-tail envelope above. Rank enumeration uses $n \leq 100000$. Actual cancellation factors are reported. Numerical values and floating-point endpoints are illustrations, not ball-arithmetic certificates.

Run from the release root with `python checks/arithmetic_inverse.py`; the project dependency fallback is discovered by the script in this session, while a portable environment needs `mpmath`. Results are written to `evidence/arithmetic_inverse_checks.json`, including input/source hashes, arithmetic conventions, tested ranges, software versions, elapsed time, and the source-retrieval failure. Universal conclusions AI-1 through AI-4 rest on the local proofs above. Historical priority remains unresolved.

C Finite-prime sieve families and explicit detection of signed atoms

Checkpoint SF-DISCOVERY-100. This extends the one-prime route to a genuine finite-prime parameter family. The results below are local mathematical derivations. They do not assert historical priority. Classical orthogonal-polynomial mechanisms are identified as such.

C.1 Definition and exact coordinates

Let F be a finite set of distinct primes, let $\boldsymbol{\lambda} = (\lambda_p)_{p\in F}$, and put

$$x_n = (1-n^{-2})^2, \qquad \kappa_F(n;\boldsymbol{\lambda}) = \prod_{p\in F}(1+\lambda_p \mathbf{1}_{p|n}),$$
$$\nu_{F,\boldsymbol{\lambda}} = \sum_{n\geq 1} \mu(n)^2 n^{-4} \kappa_F(n;\boldsymbol{\lambda})\delta_{x_n}.$$

The mass at $x_1 = 0$ is exactly 1. A strength $\lambda_p = -1$ deletes every squarefree atom divisible by p. Several deletions commute. The depth-k family uses $x^k\, d\nu$, for integers $k \geq 0$.

Write $D(s) = \zeta(s)/\zeta(2s)$. For arbitrary finite complex strengths and $\Re s > 1$, absolute convergence permits expansion of the finite prime product. The local squarefree Euler factors give

$$Z_F(s;\boldsymbol{\lambda}) = \sum_{n\geq 1} \mu(n)^2 n^{-s} \kappa_F(n;\boldsymbol{\lambda}) = D(s) \prod_{p\in F}\left(1+\frac{\lambda_p}{p^s+1}\right).$$

Equivalently, expansion in subsets $S \subseteq F$ gives the coefficient $D(s)/\prod_{p\in S}(p^s+1)$ for $\prod_{p\in S}\lambda_p$. This is an independently useful coordinate formula for prime overlap.

Consequently

$$M_j(\boldsymbol{\lambda}) = \int x^j\, d\nu_{F,\boldsymbol{\lambda}} = \sum_{r=0}^{2j}(-1)^r \binom{2j}{r} D(4+2r) \prod_{p\in F}\left(1+\frac{\lambda_p}{p^{4+2r}+1}\right). \tag{1}$$

Depth-k moments are M_{k+j}. With $v = \pi^{-2}$, the even-zeta formula makes every $D(4+2r)$ a rational multiple of v^{r+2}. Thus the moments belong to $\mathbb{Q}[v, \lambda_p : p \in F]$, and monic orthogonal-polynomial coefficients belong to $\mathbb{Q}(v, \lambda_p : p \in F)$. This is an exact field statement, with no decimal recognition step.

For two primes the interaction term is explicit:

$$M_j(\lambda_p,\lambda_q) = M_j(0,0) + \lambda_p U_{p,j} + \lambda_q U_{q,j} + \lambda_p\lambda_q U_{pq,j},$$

where $U_{pq,j} = \sum_{pq|n} \mu(n)^2 n^{-4} x_n^j > 0$. More generally, with all other strengths at least -1, every mixed moment derivative in a set of distinct primes is strictly positive. It is the integral over atoms divisible by those primes, with the remaining nonnegative prime factors. Infinitely many such atoms remain after any finite collection of deletions. Pure second derivatives in any one strength vanish because the moments are separately affine.

C.2 The exact positivity domain

Theorem 1. For real strengths, $\nu_{F,\boldsymbol{\lambda}}$ is a positive measure if and only if $\lambda_p \geq -1$ for every $p \in F$. On this closed orthant, every finite Hankel matrix of every nonnegative integer depth is strictly positive definite.

Proof. The stated inequalities make each atom weight nonnegative. The primes outside F supply infinitely many positive atoms, even when all strengths are deletion endpoints. A nonzero polynomial cannot vanish on infinitely many distinct nodes; therefore its squared norm is strictly positive. Conversely, if $\lambda_p < -1$, the squarefree atom $n = p$ has mass $(1+\lambda_p)p^{-4} < 0$: no other strength appears at that atom. The same argument works after every depth tilt because $x_p > 0$. □

The positive monic families $P_m^{(k;F,\lambda)}$ consequently exist at all degrees throughout the closed orthant. Their Hankel determinants never vanish there, including at simultaneous deletion endpoints. Each finite-degree coefficient is analytic in a neighborhood of every point of that orthant. This neighborhood is a finite-degree statement; the next theorem explains why no common enlargement of the positivity domain works at all degrees.

Together, Theorems 1 and 2 also prove the exact equivalence: every finite Hankel matrix is positive semidefinite if and only if every strength is at least -1. Thus this is a complete positivity-domain result, not only a sufficient condition.

C.3 An explicit polynomial detector for every bad strength

For an integer $a \geq 2$, define

$$L_a(x) = \prod_{n=1}^{a-1} \frac{x - x_n}{x_a - x_n}, \qquad C_a = L_a(1) = \prod_{n=1}^{a-1} \frac{1 - x_n}{x_a - x_n} > 0,$$

$$r_a = \frac{1 - x_{a+1}}{1 - x_a} = \frac{a^4\big(2(a+1)^2 - 1\big)}{(a+1)^4(2a^2 - 1)} \in (0,1),$$

$$f_{a,m}(x) = L_a(x)\left(\frac{1-x}{1-x_a}\right)^m, \qquad m = 0,1,2,\ldots. \tag{2}$$

The degree is $a - 1 + m$; all lower nodes are killed exactly; $f_{a,m}(x_a) = 1$. At every higher node, $|f_{a,m}(x_n)| \leq C_a r_a^m$, because each factor of L_a increases on $[x_a, 1]$ and $1 - x_n \leq 1 - x_{a+1}$.

Set $C_F = \prod_{q\in F} \max(1, |1+\lambda_q|)$. Then $|\kappa_F(n)| \leq C_F$, and the elementary decreasing-series bound gives

$$\sum_{n>a} |\mu(n)^2 n^{-4} \kappa_F(n)| \leq \frac{C_F}{3a^3}. \tag{3}$$

Theorem 2. If $\lambda_p < -1$, then

$$\int f_{p,m}(x)^2 \, d\nu_{F,\lambda}(x) \leq \frac{1+\lambda_p}{p^4} + \frac{C_F C_p^2}{3p^3} r_p^{2m}. \tag{4}$$

Hence the Hankel matrix of order $p + m$ is not positive semidefinite whenever

$$\boxed{\frac{pC_F C_p^2}{3} r_p^{2m} < -(1+\lambda_p).} \tag{5}$$

All quantities in (5) are rational when the strengths are rational. It provides an exact finite certificate without calculating Hankel eigenvalues or relying on floating-point root finding. A suitable finite m exists because $r_p < 1$.

Proof. The polynomial kills all atoms below p, equals 1 at the negative atom p, and has the stated uniform bound above p. Replacing all higher signed contributions by their absolute values and applying (3) proves (4). The coefficients of $f_{p,m}$ then supply a negative quadratic form for $[M_{i+j}]_{i,j<p+m}$ when (5) holds. □

For depth k, the same polynomial satisfies

$$\int x^k f_{p,m}(x)^2 \, d\nu \le \frac{(1+\lambda_p)x_p^k}{p^4} + \frac{C_F C_p^2}{3p^3} r_p^{2m}.$$

Replace the right side of (5) by $-(1+\lambda_p)x_p^k$ to obtain an explicit depth-k detector. This proof extends to any finite signed atomic measure with these nodes and the same summable weight envelope. Its mechanism is an isolating polynomial followed by exponential suppression of the higher nodes.

C.4 Conditional prime fibers and all-prime margin bounds

Fix $p \in F$, with all other strengths at least -1, and write $\kappa(n) = \prod_{q \in F\setminus\{p\}} (1+\lambda_q \mathbf{1}_{q|n})$. Separate the positive measure into its p-free and p-divisible parts:

$$A_m = [\sum_{p\nmid n} \mu(n)^2 n^{-4} \kappa(n) x_n^{i+j}]_{i,j<m}, \quad B_m = [\sum_{p|n} \mu(n)^2 n^{-4} \kappa(n) x_n^{i+j}]_{i,j<m}.$$

Both matrices are positive definite. For A_m, use infinitely many primes outside F; for B_m, use infinitely many squarefree integers $p\ell$, with primes ℓ outside F. This remains true at every deletion endpoint of the other strengths.

Theorem 3. The one-variable fiber

$$\tau_m(\lambda_p) = \det\big(A_m + (1+\lambda_p)B_m\big)$$

has exact degree m, and all its roots, with multiplicity, are real and strictly less than -1. The root nearest the positivity boundary is $-1-\delta_m$, with

$$\delta_m = \min_{0\neq f\in\mathbb{R}[x],\ \deg f<m} \frac{\int f^2 \, d\nu_{p\text{-free}}}{\int f^2 \, d\nu_{p\text{-divisible}}} > 0. \tag{6}$$

The sequence δ_m is nonincreasing, and, with $C_\kappa = \prod_{q\neq p} \max(1, 1+\lambda_q)$,

$$\boxed{\delta_{p+m} \le \frac{pC_\kappa C_p^2}{3} r_p^{2m}.} \tag{7}$$

Proof. The definite pencil is congruent to $I + (1+\lambda_p)A_m^{-1/2} B_m A_m^{-1/2}$, whose eigenvalues have positive slopes. Its roots are $-1-1/\eta_i$, with $\eta_i > 0$. This proves the degree and root statements, and the reciprocal largest eigenvalue is exactly (6). Polynomial-space inclusion proves monotonicity. Test (6) at order $p+m$ with (2). The denominator includes the atom p, of mass p^{-4}, because $\kappa(p) = 1$. The numerator kills every lower atom, has no atom at p, and its higher tail is bounded by $C_\kappa C_p^2 r_p^{2m}/(3p^3)$. This proves (7) and therefore $\delta_m \to 0$. □

A sharper arithmetic bound uses the exact remaining tail mass

$$A_{\text{tail}} = D(4)\frac{p^4}{p^4+1}\prod_{q\neq p}\left(1+\frac{\lambda_q}{q^4+1}\right) - \sum_{1\le n<p}\mu(n)^2 n^{-4}\kappa(n) > 0,$$

and replaces (7) by $\delta_{p+m} \le p^4 C_p^2 A_{\text{tail}} r_p^{2m}$. Every finite-degree coefficient can have poles only at the fiber's Hankel roots; numerator cancellations can remove a possible pole. The theorem does not assert simple roots or global multivariate real stability.

At depth zero, the initial margin is exactly $\delta_1 = p^4$, independently of all other safe strengths: the total p-free and p-divisible masses have ratio p^4, since all the other Euler factors cancel. The higher-order margins contain the arithmetic overlap information.

The same argument applies to every nonnegative depth, replacing each integral by its x^k tilt. The denominator's explicit atom contribution becomes $p^{-4}x_p^k$, and (7) acquires the factor x_p^{-k}.

C.5 Resolution, nonconvex finite tests, and a different additive family

For $F = \{2,3\}$, the order-one Hankel matrix is the scalar

$$M_0(\lambda_2,\lambda_3) = 105\pi^{-4}\left(1+\frac{\lambda_2}{17}\right)\left(1+\frac{\lambda_3}{82}\right).$$

Thus $(0,0)$ and $(-34,-164)$ have the same positive order-one Hankel matrix. Their midpoint $(-17,-82)$ has zero total mass. The unrestricted real parameter set passing the order-one positive-definiteness test is therefore nonconvex. This is a concrete warning about a finite test, while the true positivity domain in Theorem 1 is a convex closed orthant.

The positive-semidefinite feasible set is nonconvex as well: the points $(0,0)$ and $(-34,-246)$ have positive mass factors 1 and 2, while their convex combination with weights $3/5$ and $2/5$, namely $(-68/5,-492/5)$, has mass factor $-1/25$.

An additive prime family is a different deformation:

$$\widetilde{\nu}_{F,\gamma} = \sum_{n\ge 1}\mu(n)^2 n^{-4}\left(1+\sum_{p\in F}\gamma_p \mathbf{1}_{p|n}\right)\delta_{x_n}.$$

Its exact positivity domain is the polyhedron

$$1+\sum_{p\in S}\gamma_p \ge 0 \quad (S\subseteq F), \quad \text{equivalently} \quad 1+\sum_{\gamma_p<0}\gamma_p \ge 0.$$

Necessity uses the squarefree integer $n = \prod_{p\in S} p$; sufficiency checks every atom's prime subset. Its finite Hankel matrices are affine functions of the strengths, so their positive-semidefinite feasible sets are convex spectrahedra. In contrast, the multiplicative family includes the arithmetic prime-overlap terms and has polynomial parameter dependence. For example, additive strengths $\gamma_2 = \gamma_3 = -3/4$ leave both individual prime weights positive but make the $n = 6$ weight negative; multiplicative strengths $\lambda_2 = \lambda_3 = -3/4$ remain safely positive.

C.6 A limited nonreduction statement

If at least two strengths in a positive multiplicative family are nonzero, its entire depth-zero monic polynomial sequence cannot equal any one-prime sequence on the same variable and node system. A direct proof avoids importing a determinacy theorem. Write $P_m = x^m + \sum_{j<m} c_{mj} x^j$. The equations $\int P_m \, d\nu = 0$ recursively determine every normalized moment from the monic sequence. Equal sequences therefore give equal normalized Stieltjes series for $|z| > 1$. Analytic continuation on the connected complement of the common closed node set makes their Stieltjes functions equal, and residues at each isolated node give equal normalized atom weights. The mass 1 at zero fixes the normalization constant. Finally, the atom at each prime p reads off $1 + \lambda_p$, whereas a one-prime deformation changes at most one such atom. This is a precise separator from the existing one-prime family, not a global novelty certificate. It does not address arbitrary affine variable changes.

The multi-prime Stieltjes kernel also retains infinitely many isolated poles at primes outside F, so the preceding one-prime proof of non-D-finiteness applies unchanged. Orthogonal-polynomial existence, definite-pencil spectral theory, and compact moment determinacy are classical mechanisms. The specific arithmetic coordinates, parameter geometry, and explicit finite-resolution detector are the objects developed here.

C.7 Worked arithmetic examples

For $F = \{2, 3\}$, the following coefficients are rounded high-precision evaluations of (1) and the Hankel formulas. They are not exact decimal constants.

(λ_2, λ_3)	$P_1(x)$	$P_2(x)$
$(-1, -1)$	$x - 0.00200014863265043361$	$x^2 - 0.933902261393426053x + 0.000000921271332319859170$
$(-1/2, 1)$	$x - 0.0377167661732468061$	$x^2 - 0.725293001442481628x + 0.00093264814608580611 2$
$(1, 1)$	$x - 0.0822907496962840844$	$x^2 - 0.655873200757636623x + 0.00211153000026086129$

Simultaneously deleting $2, 3, 5$ gives $P_1(x) = x - 0.0005287214220477823 75$ and $P_2(x) = x^2 - 0.966491641778214 72x + 0.000000079553141379691 0514$, again rounded. This removes the low prime shelves while retaining the fixed mass at zero and an infinite tail.

For the variable strength at $p = 2$, the nearest conditional margins are:

Hankel order	Other strength $\lambda_3 = -1$	Other strength $\lambda_3 = 3$
1	16, exact	16, exact
2	0.0939109919936561	1.43829172807931
3	0.000302450514004102	0.0346655628067259
4	0.000000528145808944195	0.000223749375519984

The noninteger entries are approximate symmetric-pencil calculations. They show an actual prime-overlap effect beyond the mass test. In these calculations the defect $\lambda_2 = -1.001$ is visible at order 3 when the 3-shelf is deleted, but only at order 4 when that shelf is strengthened to 3. This comparison is a numerical example, not a universal monotonicity theorem.

The separate rational certificates (5) do not depend on those approximate roots. The script constructs negative detectors for primes $2, 3, 5, 7$, at Hankel orders $8, 11, 19, 40$, respectively. These sufficient orders are not asserted to be minimal. In the $p = 7$ example, the other strengths are $\lambda_2 = 1/2, \lambda_3 = -1, \lambda_5 = 2$, and $\lambda_7 = -1.0001$; the exact rational upper bound on the detector norm is approximately -9.74271×10^{-9}.

C.8 Computation contract

`checks/multi_prime_sieve.py` uses exact rational arithmetic for Euler-factor versus subset-expansion comparisons, Bernoulli-based even-zeta coordinates, isolating-polynomial values, and the rational detection criterion (5). It separately evaluates the infinite moments at high precision, tests positive orthogonality grids, compares them with a direct finite squarefree sum plus a proved tail envelope, calculates conditional definite-pencil roots, and evaluates signed detector norms. These decimal checks are not interval certificates. Universal positivity and negativity conclusions follow from the displayed proofs and exact rational bounds.

The passing run records 1,019 exact rational assertions, 102 positive parameter/depth cases, 1,395 lower-monomial orthogonality conditions, 381 recurrence identities, 40 conditional-pencil cases, and four exact negative-atom certificates. Working precision is 320 decimal digits. Maximum absolute orthogonality residual was approximately 6.33×10^{-322}, maximum recurrence residual 2.23×10^{-303}, and maximum scale-normalized pencil determinant residual 5.06×10^{-321}. These are finite numerical consistency results, not interval certificates. No execution failure occurred in this route's run.

The script records actual versions, source hash, parameter bounds, finite counts, precision, residuals, examples, and any failed criteria in `evidence/multi_prime_sieve_checks.json`. It accepts an optional output path so a parent bounded runner can place the result inside a fresh evidence directory. No broad priority search was performed for this specific extension; historical novelty remains unresolved.

D Orbit spectral zeta: continuation, trace zeros, and regularized product

Contract: SF-DISCOVERY-100-SPECTRAL. Attribution: Komorebi P. This extends the v0.1 orbit family with a self-contained continuation proof. The Mittag-Leffler seed and Mellin continuation mechanism are classical. Historical priority for this exact specialization is not established.

D.1 The completed result

Let $N \geq 2$ be an integer, and put $\alpha = 1/N$. Define

$$d_N(w) = \sum_{n=0}^{\infty} \frac{(-1)^n w^n}{(Nn)!} = E_{N,1}(-w).$$

List its nonzero zeros w_j with multiplicity, and use the principal logarithm, with $-\pi < \operatorname{Arg} w_j < \pi$. No zero lies on the omitted negative ray. The orbit zeta function

$$\mathcal{Z}_N(s) = \sum_j e^{-s \operatorname{Log} w_j}$$

converges absolutely exactly when $\Re s > 1/N$. It has a meromorphic continuation to the whole plane, with exactly one pole: a simple pole at $s = 1/N$, of residue

$$\boxed{\operatorname{Res}_{s=1/N} \mathcal{Z}_N(s) = \frac{\sin(\pi/N)}{N\pi}.}$$

At every nonpositive integer,

$$\boxed{\mathcal{Z}_N(-m) = 0 \quad (m = 0, 1, 2, \ldots).}$$

At the origin,

$$\boxed{\mathcal{Z}_N'(0) = -\log N, \qquad \exp[-\mathcal{Z}_N'(0)] = N.}$$

The last equality is the zeta regularized product of the orbit zeros under this branch convention. It is a regularized product, not convergence of the ordinary product of the zeros. Positive integer values retain the exact rational coefficient law from v0.1.

D.2 The exponential packet and its exact asymptotic

Let $\omega = e^{2\pi i/N}$. The finite root-of-unity filter gives, for every $x \in \mathbb{C}$,

$$F_N(x) := d_N(-x^N) = \frac{1}{N} \sum_{j=0}^{N-1} e^{\omega^j x} = \sum_{n=0}^{\infty} \frac{x^{Nn}}{(Nn)!}. \tag{S1}$$

This equality follows by expanding the exponentials and using $N^{-1} \sum_j \omega^{jm} = 1$ when $N \mid m$, and zero otherwise. For real $x \geq 0$, the power series makes $F_N(x) > 0$. Put

$$h_N(x) = \sum_{j=1}^{N-1} e^{(\omega^j - 1)x}, \qquad c_N = 1 - \cos(2\pi/N) > 0.$$

Then

$$F_N(x) = \frac{e^x}{N}(1 + h_N(x)), \qquad |h_N(x)| \le (N-1)e^{-c_N x}, \qquad |h_N'(x)| \le 2(N-1)e^{-c_N x}. \tag{S2}$$

Since $1 + h_N(x) = Ne^{-x}F_N(x) > 0$, the real logarithm is well defined for all $x \ge 0$. At infinity,

$$\log F_N(x) = x - \log N + O(e^{-c_N x}), \qquad g_N(x) := \frac{F_N'(x)}{F_N(x)} = 1 + O(e^{-c_N x}). \tag{S3}$$

At the origin $g_N(x) = O(x^{N-1})$. Its only Taylor powers are x^{Nm-1}, with $m \ge 1$.

D.3 The branch and absolute-interchange issue

The pre-existing genus-zero product is

$$d_N(w) = \prod_j (1 - w/w_j). \tag{S4}$$

For completeness, the series bound $|d_N(w)| \le \exp(|w|^{1/N})$ gives order at most $1/N < 1$. Hadamard factorization therefore has genus zero and a constant exponential factor, which equals one because $d_N(0) = 1$. The usual zero-count consequence gives $\sum_j |w_j|^{-\sigma} < \infty$ for every $\sigma > 1/N$. Along the negative real axis (S3) also shows the order is exactly $1/N$.

It is not enough simply to say that no zero lies on the integration contour. We need a uniform angular gap from the branch cut to interchange the Mellin integral with the zero sum. Here is that gap.

For $|\arg(-w)| \le \pi/2$, choose $x = (-w)^{1/N}$ with $|\arg x| \le \pi/(2N)$. For every $j \ne 0$,

$$\Re x - \Re(\omega^j x) \ge \left[\cos\frac{\pi}{2N} - \cos\frac{3\pi}{2N}\right] |x| =: \gamma_N |x| > 0. \tag{S5}$$

Indeed, the nearest competing root is separated from the positive direction by at least $3\pi/(2N)$, while the distinguished root is within $\pi/(2N)$. Cosine decreases on $[0, \pi]$. Thus, once $(N-1)e^{-\gamma_N|x|} < 1$, the distinguished term e^x in (S1) is larger in modulus than the sum of all competitors, and $d_N(w) \ne 0$. Consequently, sufficiently large zeros avoid the entire closed left half-plane. There are only finitely many remaining zeros in a bounded disk. None lies on the negative ray, since $d_N(-t) > 0$ for $t \ge 0$. It follows that some $\eta_N > 0$ satisfies

$$|\operatorname{Arg} w_j| \le \pi - \eta_N \qquad \text{for every } j. \tag{S6}$$

With $c = \sin(\eta_N/2) > 0$, this implies

$$|t + w_j| \ge c(t + |w_j|), \qquad t \ge 0. \tag{S7}$$

To check the elementary estimate, write $w = re^{i\theta}$: $|t + w|^2 = (t + r)^2 - 4tr\sin^2(\theta/2) \ge (t + r)^2\cos^2(\theta/2)$.

D.4 Mellin representation in the original strip

Let $H_N(t) = d_N(-t) = E_{N,1}(t)$, and $G_N(t) = H_N'(t)/H_N(t)$. Logarithmic differentiation of (S4), locally uniformly off the zeros, gives

$$G_N(t) = \sum_j \frac{1}{t + w_j}, \qquad t \geq 0. \tag{S8}$$

For $1/N < \sigma < 1$, (S7) yields

$$\sum_j \int_0^\infty \frac{t^{-\sigma}}{|t + w_j|}\, dt \leq \frac{\pi}{c \sin(\pi\sigma)} \sum_j |w_j|^{-\sigma} < \infty. \tag{S9}$$

This is the absolute interchange needed for complex s with $\Re s = \sigma$. Euler's beta integral and the reflection identity give

$$\int_0^\infty \frac{t^{-s}}{t + w}\, dt = \frac{\pi}{\sin(\pi s)} e^{-s \operatorname{Log} w}, \quad 0 < \Re s < 1, \quad |\arg w| < \pi. \tag{S10}$$

First prove this for positive w by substitution and the beta integral, then extend to the slit w-plane by holomorphic dependence. On compact subsets of that slit plane the denominator estimate supplies the needed local domination.

Combining (S8)–(S10), and substituting $t = x^N$, proves

$$\boxed{\mathcal{Z}_N(s) = \frac{\sin(\pi s)}{\pi} \int_0^\infty x^{-Ns} g_N(x)\, dx, \qquad \frac{1}{N} < \Re s < 1.} \tag{S11}$$

For $x > 0$, $x^{-Ns} = e^{-Ns \log x}$ uses the real logarithm. Multiplicities are already incorporated in (S4), so no simple-zero assumption entered the proof.

D.5 Global continuation and the only pole

One can see directly that d_N has no zero on $|w| \leq 1$:

$$|d_N(w) - 1| \leq \sum_{n \geq 1} \frac{1}{(Nn)!} \leq \cosh(1) - 1 < 1.$$

The strict inequality also gives a slightly larger zero-free disk by continuity. Thus $G_N(t)$ has a Taylor series converging uniformly on $[0, 1]$,

$$G_N(t) = \sum_{k \geq 0} a_{N,k} t^k, \qquad g_N(x) = \sum_{m \geq 1} N a_{N,m-1} x^{Nm-1}. \tag{S12}$$

For any integer $K \geq 0$, define

$$B_{N,K}(s) = \int_0^1 x^{-Ns}\left[g_N(x) - \sum_{m=1}^{K} N a_{N,m-1} x^{Nm-1}\right] dx + \sum_{m=1}^{K} \frac{a_{N,m-1}}{m-s} + \int_1^\infty x^{-Ns}\big[g_N(x)-1\big]dx + \frac{1}{Ns-1}. \tag{S13}$$

The first integral is holomorphic for $\Re s < K+1$. This includes local uniform domination, since the Taylor remainder is $O(x^{N(K+1)-1})$. The last integral is entire in s, because (S3) gives exponential decay against every power of x, uniformly on compact s-sets, including after any number of s-derivatives. The rational terms exhibit the only possible poles: positive integers $1,\dots,K$ and $1/N$.

On their overlapping domains the functions (S13) agree: adding the next Taylor term subtracts its integral and adds precisely the same rational function. They also agree with (S11) in its original strip. Therefore

$$\mathcal{Z}_N(s) = \frac{\sin(\pi s)}{\pi} B_{N,K}(s) \tag{S14}$$

provides compatible meromorphic continuations covering the whole plane as $K \to \infty$. At each positive integer m, the apparent simple pole $a_{N,m-1}/(m-s)$ cancels with the simple zero of $\sin(\pi s)$. The term $1/(Ns-1)$ has residue $1/N$, and its multiplier at $s = 1/N$ is nonzero. This proves the stated unique simple pole and its residue.

It also proves that the original series cannot converge absolutely at $s = 1/N$. If it did, its absolute terms would dominate the series for real $s \downarrow 1/N$, except for finitely many zeros of modulus below one. Dominated convergence would then keep that real limit bounded, contradicting the nonzero pole. Absolute convergence at any smaller real part would imply convergence at $1/N$. The abscissa is consequently exactly $1/N$.

D.6 Trace zeros, rational values, and the origin

For $\Re s < 1$ we may use $K = 0$:

$$B_{N,0}(s) = \int_0^1 x^{-Ns} g_N(x)\, dx + \int_1^\infty x^{-Ns}(g_N(x)-1)\, dx + \frac{1}{Ns-1}. \tag{S15}$$

This expression is holomorphic at every nonpositive integer. Hence (S14) proves $\mathcal{Z}_N(-m) = 0$. These statements do not assert that every such zero is simple.

At a positive integer m, taking the removable limit in (S14) gives

$$\mathcal{Z}_N(m) = (-1)^{m+1} a_{N,m-1}. \tag{S16}$$

If $c_n = (-1)^n/(Nn)!$, the previous logarithmic-product law therefore remains

$$b_m := \mathcal{Z}_N(m) = -m c_m - \sum_{j=1}^{m-1} c_j b_{m-j}. \tag{S17}$$

In particular all these values are rational. This is the finite coefficient dictionary, independent of numerical zero location.

At $s = 0$, the elementary antiderivatives in (S15) give

$$B_{N,0}(0) = \log F_N(1) + \lim_{R\to\infty}\big[\log F_N(R) - R - \log F_N(1) + 1\big] - 1$$
$$= -\log N.$$

Since $\sin(\pi s)/\pi = s + O(s^3)$, (S14) implies $\mathcal{Z}_N(0) = 0$ and $\mathcal{Z}_N'(0) = -\log N$.

D.7 Classical benchmarks and numerical examples

The inherited identities

$$\mathcal{Z}_2(s) = \pi^{-2s}\zeta(2s, 1/2), \qquad \mathcal{Z}_4(s) = (\sqrt{2}\pi)^{-4s}\zeta(4s, 1/2)$$

now hold as meromorphic identities, by equality in the original half-plane and uniqueness of continuation. Their pole residues are respectively $1/(2\pi)$ and $1/(4\sqrt{2}\pi)$, agreeing with the general theorem. Their zero values at nonpositive integers follow from the Hurwitz formula, and their regularized products are respectively 2 and 4.

The companion script performs 411 exact rational assertions: independent formal-logarithm and quotient calculations for $2 \le N \le 10$ through degree 40; three explicit integer values for each N; and 12 Bernoulli/Hurwitz anchors each for $N = 2, 4$. It then evaluates (S13) using a truncated local Taylor sum and high-precision exponential-tail quadrature. Sixteen $N = 2, 4$ benchmarks sample both sides of the original convergence boundary, negative real values, and complex values. Values for $N = 3, 5$ and checks of $\mathcal{Z}_N'(0)$ are recorded in the JSON evidence.

Those quadrature results are approximations. Agreement at two precisions and with known functions tests normalization and implementation; it does not replace the proof, certify intervals, or settle historical novelty.

D.8 A worked Hurwitz deformation

There is a concrete shifted family. For a real shift $a \ge 0$, define

$$\mathcal{Z}_N(s; a) = \sum_j (w_j + a)^{-s}, \qquad \Re s > 1/N, \tag{T1}$$

again with the principal logarithm and multiplicities. A rightward shift preserves the angular gap from the negative ray: for each nonreal w, $|\arg(w + a)|$ is nonincreasing in a, and positive real zeros remain positive. The zero sequence is unchanged at infinity up to a bounded translation, so the previous absolute convergence argument applies. The normalized determinant is exactly

$$\frac{H_N(a+t)}{H_N(a)} = \prod_j \left(1 + \frac{t}{w_j + a}\right), \qquad H_N(t) = E_{N,1}(t) > 0 \quad (t \ge 0). \tag{T2}$$

This follows by taking the ratio of the two convergent products in (S4), not by introducing an unverified exponential factor. Its logarithmic derivative is $G_{N,a}(t) = H_N'(a+t)/H_N(a+t)$. The same dominated beta-integral argument gives

$$\mathcal{Z}_N(s; a) = \frac{\sin(\pi s)}{\pi} \int_0^\infty t^{-s} G_{N,a}(t)\, dt, \qquad 1/N < \Re s < 1. \tag{T3}$$

At infinity, (S2)–(S3), with $x = (a+t)^{1/N}$, imply

$$G_{N,a}(t) = \alpha(t+a)^{\alpha-1} + O\Big((t+a)^{\alpha-1}e^{-c_N(t+a)^{\alpha}}\Big). \tag{T4}$$

For fixed a and any integer $K \geq 0$, the binomial expansion therefore gives

$$G_{N,a}(t) = \sum_{k=0}^{K} \alpha \binom{\alpha-1}{k} a^k t^{\alpha-k-1} + O(t^{\alpha-K-2}). \tag{T5}$$

To be explicit about continuation, choose a finite splitting point $T > 2a+1$. Subtract the first $K+1$ terms of (T5) inside the integral from T to infinity, and add their integrals back as the rational terms

$$\frac{\alpha\binom{\alpha-1}{k}a^k T^{\alpha-k-s}}{s-\alpha+k}. \tag{T6}$$

The remainder integral is holomorphic for $\Re s > \alpha - K - 1$. The integral on any compact interval away from zero is entire. At zero, $G_{N,a}$ is analytic since $H_N(a) > 0$; subtracting any finite Taylor polynomial there extends the other side, with apparent poles only at positive integers. Such Taylor poles cancel against $\sin(\pi s)$, just as in Section 4. Choosing arbitrarily large subtractions at both ends covers every $s \in \mathbb{C}$. The resulting continuations agree on overlaps because all added terms are exactly the integrals of the terms removed.

For $a > 0$, the complete pole set is consequently

$$\boxed{s = \frac{1}{N} - k, \quad k = 0, 1, 2, \ldots,}$$

and every pole is simple. Its residue is

$$\begin{aligned} \operatorname{Res}_{s=\alpha-k} \mathcal{Z}_N(s;a) &= \frac{\alpha \sin(\pi(\alpha-k))}{\pi} \binom{\alpha-1}{k} a^k \\ &= \boxed{\frac{\sin(\pi/N)}{N\pi} \frac{(1-1/N)_k}{k!} a^k.} \end{aligned} \tag{T7}$$

Here $(b)_0 = 1$ and $(b)_k = b(b+1)\cdots(b+k-1)$. The sign calculation uses both $\sin(\pi(\alpha-k)) = (-1)^k \sin(\pi\alpha)$ and $\binom{\alpha-1}{k} = (-1)^k(1-\alpha)_k/k!$; their signs cancel. All residues are positive when $a > 0$. At $a = 0$, the secondary coefficients vanish, so the only remaining pole is the unshifted pole at $1/N$. As before, its nonzero residue makes the absolute convergence abscissa exactly $1/N$.

None of the pole locations $\alpha - k$ is an integer. Thus the sine factor vanishes at every nonpositive integer without an opposing pole:

$$\boxed{\mathcal{Z}_N(-m;a) = 0 \quad (m \geq 0).} \tag{T8}$$

Finally, the one-term subtraction is already convergent at $s = 0$, since $G_{N,a}(t) - \alpha t^{\alpha-1} = O(t^{\alpha-2})$ and $\alpha < 1$. Let

$$B_{N,a}(0) = \int_0^1 G_{N,a}(t)dt + \int_1^{\infty} [G_{N,a}(t) - \alpha t^{\alpha-1}]dt - 1.$$

Integrating exactly and using $\log H_N(a+t) - t^\alpha \to -\log N$ gives

$$B_{N,a}(0) = -\log N - \log H_N(a).$$

The multiplier $\sin(\pi s)/\pi$ then gives

$$\boxed{\mathcal{Z}'_N(0;a) = -\log[NE_{N,1}(a)], \qquad \exp[-\mathcal{Z}'_N(0;a)] = NE_{N,1}(a).} \tag{T9}$$

For $N = 2$, the product reduces to $2\cosh\sqrt{a}$. Four direct quadrature checks at $a = 0, 1/4, 2, 9$ agree with $-\log(2\cosh\sqrt{a})$ at 80-digit working precision. They use a local integral, an exactly integrated algebraic leading tail, and an exponentially decaying remainder; the quadrature does not sample numerical zero locations. These are four additional numerical benchmarks, with the original 411 exact assertions unchanged. Results are in `evidence/spectral_continuation_checks.json` under `shifted_N2_product_benchmarks`.

This is a classical Hurwitz-type spectral deformation, with a packet-specific explicit regularized determinant. It is useful because a single shift reveals an entire pole ladder that collapses at the unshifted boundary. No historical-priority claim is made for the deformation or determinant identity.

D.9 Source comparison and bounded search

Checked 8 October 2026 UTC. No external global-continuation theorem is required by the local proof above beyond the already authenticated finite-order Hadamard product and the elementary beta/reflection identities. The Mellin subtraction mechanism itself is established.

- NIST DLMF 10.46.3: the definition of $E_{a,b}$ identifies this seed with the classical Mittag-Leffler function. Source: `https://dlmf.nist.gov/10.46`.
- NIST DLMF 5.12.1 and 5.5.3: Euler's beta integral and the gamma reflection identity supply (S10), with the project's exponents obtained by $a = 1 - s, b = s$. Sources: `https://dlmf.nist.gov/5.12` and `https://dlmf.nist.gov/5.5`. The extension to complex w and the uniform branch-gap argument are supplied locally above.
- Noam Elkies, finite-order-function teaching notes, first Hadamard theorem: supplies the genus-zero product for an entire function of order below one. Source: `https://people.math.harvard.edu/~elkies/M259.02/prod.pdf`. The project's constant factor and zero-count consequences were already verified in v0.1 and are restated in Section 2.
- Andre Voros, **Zeta functions for the Riemann zeros**, *Annales de l'Institut Fourier* 53(3) (2003), 665–699. The authenticated journal PDF, rather than an evolving preprint, was checked. Section 2.1 specifies general admissible zero sequences of order below one and logarithmic asymptotics; Section 2.2 derives global meromorphic continuation by Mellin subtraction, positive-integer cancellations, and origin trace constants. Source: `https://www.numdam.org/article/AIF_2003__53_3_665_0.pdf`. This authenticates the shared method. It does not state this particular Mittag-Leffler packet theorem verbatim. All hypotheses needed here, including the complex-zero branch and domination conditions, were established independently in Sections 1–4. Classification: **known mechanism after translation, with a worked seed-specific specialization**.
- A separate Voros preprint, `https://arxiv.org/pdf/math/0311029,` was inspected by the coordinating agent. Its displayed arXiv v1 date and body dateline differ. This report relies on the fixed 2003 journal source above for method attribution and does not infer version-specific theorem numbering from the other file.

The bounded discovery searches used both available search engines, with the exact and adjacent terms "Mittag Leffler zeros zeta function E_n,1 analytic continuation spectral zeta log n", "hyper Bessel zeta function analytic continuation Mittag Leffler zeros integer order sin pi n", "Mittag-Leffler zeta zeros spectral meromorphic", "hyper-Bessel zeta function continuation", and "E_{N,1} zeros zeta". The closest authenticated result inspected was Voros's general framework. Search results referring to the unrelated Mittag-Leffler theorem or Helson zeta were discarded. Specialist book coverage and a complete backward/forward citation search were not exhausted. No global-priority claim follows from this search.

D.10 Scope and remaining questions

The global continuation gap for this unweighted orbit family, and for its nonnegative real shifts, is closed by the local proof. A functional equation is not supplied or presumed. The only-pole theorem concerns the unshifted case; positive shifts have the full pole ladder in (T7). The pole and trace-zero theorems follow from the exact positive-axis exponential asymptotic, and do not imply an RH-type critical-line property. Residue-weighted sums still require separate multiplicity and convergence analysis. A historical comparison against all Mittag-Leffler and hyper-Bessel zero-zeta literature remains open.

No mathematical counterexample was found during this route. Any computational failure and its repair, if encountered, must remain recorded in the evidence history rather than silently erased.

E Exact sequence, polyphase and tap laboratory

Contract: **SF-DISCOVERY-100**. Author attribution: **Komorebi P.**

This chapter connects the nonlinear family to exact sequences and finite filter algebra. It supplies reusable code without replacing the analytic proofs in `routes/asymmetric_dilation.md`. All mathematical sequence indices begin at zero. All finite arrays are zero extended beyond their displayed support. The catalog contains explicit candidates for OEIS lookup. This chapter does not assert an OEIS accession or historical priority for an unsearched row.

E.1 The integer polynomial object

Write the unique odd formal solution as

$$T_{q,r}(z) = \sum_{n\ge 0} c_n(q,r) z^{2n+1} = \sum_{n\ge 0} A_n(q,r) \frac{z^{2n+1}}{(2n+1)!}.$$

The two coefficient systems are related by $A_n = (2n+1)!c_n$. In particular, A_n is the derivative of order $2n+1$ at zero. The coefficient c_n, the derivative A_n, and the full EGF sequence a_m with $a_{2n+1} = A_n, a_{2n} = 0$ are different objects. None of the code silently substitutes one for another.

The nonlinear differential equation gives the recurrence

$$\boxed{A_0 = 1, \qquad A_n = \sum_{i=0}^{n-1} \binom{2n}{2i+1} q^{2i+1} r^{2(n-1-i)+1} A_i A_{n-1-i}, \quad n \ge 1.}$$

Proof. Multiply the ordinary Taylor recurrence by $(2n+1)!$. The factor attached to $A_i A_j$, where $i+j = n-1$, is $(2n)!/[(2i+1)!(2j+1)!] = \binom{2n}{2i+1}$. This also proves the recurrence directly in the EGF convention. □

Polynomial theorem. For every n, $A_n \in \mathbb{Z}[q,r]$, all its nonzero coefficients are positive integers, and $A_n(q,r) = A_n(r,q)$. Every monomial $q^u r^v$ satisfies

$$u, v \ge n, \qquad u \equiv v \equiv n \pmod 2, \qquad u + v \le n(n+1).$$

Proof. The recurrence is an induction using positive integer binomial coefficients. Swapping q, r and replacing i by $n-1-i$ leaves the sum unchanged. If child indices are i, j, their exponent pairs receive the shifts $(2i+1, 2j+1)$. The lower bounds follow from $i+j+1 = n$. The two parities are the parity of $i+j+1$, also n. For the degree bound, a recurrence term has total degree at most

$$2n + i(i+1) + j(j+1) \le n(n+1), \qquad i+j = n-1,$$

since $i(i+1)+j(j+1)$ is largest at $(i,j) = (0, n-1)$ or its swap. The base $A_0 = 1$ satisfies every assertion. □

The first four polynomials are

$$\begin{aligned} A_0 &= 1, \\ A_1 &= 2qr, \\ A_2 &= 8q^2r^2(q^2+r^2), \\ A_3 &= q^3r^3\left[48(q^2+r^2)(q^4+r^4) + 80q^2r^2\right]. \end{aligned}$$

Consequently each integer pair (q, r) gives an integer sequence for all n, independently of whether its formal Taylor series converges. For example, the first values of $A_n(1, 2)$ are $1, 4, 160, 35200, 38041600$.

E.1.1 Rational feedback banks also admit integer sequences

For rational channel tuples (h_t, q_t, r_t), use

$$F'(z) = 1 + \sum_t h_t F(q_t z) F(r_t z), \qquad F(0) = 0.$$

The preceding recurrence is summed over channels with the factor h_t. In the universal polynomial over the weights and contractions, every term of A_n has weight degree exactly n and total contraction degree at most $n(n+1)$. The weight statement follows from one new weight at each recurrence step and $1 + i + j = n$; the contraction bound uses the same induction.

Let H be a common denominator for all channel weights, and D a common denominator for all channel contractions. Then

$$\boxed{B_n = H^n D^{n(n+1)} A_n \in \mathbb{Z} \quad \text{for all } n.}$$

This is a proved normalization, even with signed rational weights. It is usually not the smallest possible denominator clearing. The catalog records both A_n and B_n, and the exact multiplier, so that a proposed sequence lookup can state its normalization honestly.

E.2 A sharp distinction between sequences and analytic germs

The entire-function theorem uses $0 < q, r < 1$. Specializations with large positive integers produce valid formal and integer sequences but do not necessarily produce analytic functions near zero.

Zero-radius theorem. If $q > 0$ and $r > 1$, the formal series for $T_{q,r}$ has radius of convergence zero. The same holds after swapping the parameters.

Proof. Every coefficient is positive. In the ordinary recurrence keep only the term $i = 0$. This gives

$$c_n \geq \frac{q r^{2n-1}}{2n+1} c_{n-1} \quad \Longrightarrow \quad c_n \geq \frac{q^n r^{n^2}}{3 \cdot 5 \cdots (2n+1)}.$$

Since the denominator is at most $(2n+1)^n$,

$$c_n^{1/(2n+1)} \geq q^{n/(2n+1)} r^{n^2/(2n+1)} (2n+1)^{-n/(2n+1)} \longrightarrow \infty.$$

The Cauchy–Hadamard criterion therefore gives radius zero. □

More generally, a positive feedback bank with a channel satisfying $h > 0, q > 0, r > 1$ has the same obstruction, replacing q^n by $(hq)^n$. Positivity prevents cancellation. No conclusion of this form is asserted for a bank with signed weights.

Parameters or bank	First five A_n	Mathematical status
$(1, 1)$	1, 2, 16, 272, 7936	Known tangent derivatives
$(0, 1)$	1, 0, 0, 0, 0	Known linear boundary

Parameters or bank	First five A_n	Mathematical status
$(1,-1)$	$1,-2,16,-272,7936$	Known hyperbolic tangent derivatives
$(1,2)$	$1,4,160,35200,38041600$	Integer sequence, radius zero
$(1/2,1/2)$	$1,1/2,1/4,1/8,1/16$	Known scaled hyperbolic sine
$(1/4,3/4)$	$1,3/8,45/256,21465/262144,$ $39833775/1073741824$	Entire asymmetric family, order one

The complete catalog contains **18 rows**, each through $n = 12$: three classical or degenerate integer anchors, eight positive integer formal specializations, four rational contractions, one entire edge case of infinite order, and two normalized feedback banks. Integer integrality and rational clearing are established by the preceding proofs, not merely inferred from the finite sample.

E.2.1 The uncontracted boundary is entire of infinite order

Boundary theorem. For $0 < q < 1$, the normalized formal solution of

$$T'(z) = 1 + T(qz)T(z), \qquad T(0) = 0$$

extends to an entire function of infinite order. The swapped boundary has the same property.

Proof of entire extension. The tangent-coefficient majorant gives an initial holomorphic germ. Suppose T is known on $|z| < R$. The function $T(qz)$ is known on $|z| < R/q$. On this larger disk solve the linear ODE $U' = 1 + T(qz)U$, $U(0) = 0$. Explicitly,

$$U(z) = e^{B(z)} \int_0^z e^{-B(w)}\, dw, \qquad B(z) = \int_0^z T(qw)\, dw.$$

The integrals are path independent on the disk. Linear-ODE uniqueness shows $U = T$ on the previous disk, and there $T(qz)$ equals the newly extended function at qz. Thus the same nonlinear equation holds on the enlarged disk. Iterating $R \mapsto R/q$ exhausts the plane.

Proof of infinite order. Positive coefficients imply $T(x) \geq x$ for $x > 0$ and $M(x) = T(x)$. The logarithmic derivative obeys

$$\frac{d}{dx} \log T(x) = \frac{1}{T(x)} + T(qx) \geq qx.$$

Integrating from one, with $T(1) \geq 1$, gives $T(x) \geq \exp(q(x^2-1)/2)$ for $x \geq 1$. For $R \geq 2/q$, integrate the same logarithmic derivative over $[R/2, R]$. Since $T(R/2) \geq 1$,

$$\boxed{\log M(R) \geq \frac{R}{2} \exp\left(\frac{q^3R^2}{8} - \frac{q}{2}\right).}$$

Therefore $\log\log M(R)/\log R \to \infty$, which proves infinite order. □

For positive parameters this gives a complete analytic/formal phase chart:

Parameter region	Normalized solution
$0 < q, r < 1$	Entire, finite order defined by $q^\rho + r^\rho = 1$
One parameter equals one, the other lies in $(0, 1)$	Entire, infinite order
$q = r = 1$	Tangent, meromorphic with its usual poles
$q > 0, r > 0, \max(q, r) > 1$	Formal solution with radius zero
One parameter equals zero	Linear solution z, regardless of the other scale

The finite-order formula applies in the open contraction square. It does not assign a finite order to the edge, where its defining scale equation has no positive root.

E.3 Exact root-filter sections

For an EGF $G(z) = \sum_{m\ge 0} a_m z^m/m!$, an integer base $b \ge 1$, a residue $0 \le j < b$, and $\omega = e^{2\pi i/b}$, define

$$\boxed{\mathcal{P}_{b,j}G(z) = \frac{1}{b}\sum_{k=0}^{b-1} \omega^{-jk} G(\omega^k z) = \sum_{m\equiv j\,(b)} a_m \frac{z^m}{m!}.}$$

This holds as a formal series and wherever the original series converges. Indeed the coefficient multiplier is the geometric sum $b^{-1}\sum_{k=0}^{b-1}\omega^{k(m-j)}$, equal to one or zero according to the residue. It proves all the projector identities

$$\sum_{j=0}^{b-1} \mathcal{P}_{b,j} = I, \qquad \mathcal{P}_{b,j}\mathcal{P}_{b,l} = \delta_{jl}\mathcal{P}_{b,j}.$$

The full-position section and the decimated lane are distinct arrays. A section keeps the original positions, inserting zeros outside the residue. A lane $x_j[n] = a_{bn+j}$ changes the index to n.

For an **ordinary** generating function $X(z) = \sum_m a_m z^m$, define $X_j(w) = \sum_n a_{bn+j} w^n$. Then

$$X(z) = \sum_{j=0}^{b-1} z^j X_j(z^b).$$

For an EGF the corresponding exact reconstruction is

$$G(z) = \sum_{j=0}^{b-1} z^j \sum_{n\ge 0} \frac{a_{bn+j}}{(bn+j)!}(z^b)^n.$$

One must retain $(bn+j)!$. Replacing it by $n!$ gives a different function. The root-filter section identity has no such factorial ambiguity.

E.4 Ordinary taps, EGF multiplication and differentiation

Let $h_0, \dots, h_{L-1}$ be a finite tap vector and let $a_m = 0$ for $m < 0$. Ordinary taps form

$$b_m = \sum_{t=0}^{L-1} h_t a_{m-t}.$$

For OGFs this is multiplication by $H(z) = \sum_t h_t z^t$: $B(z) = H(z)X(z)$. For EGFs it is a sum of repeated zero-constant antiderivatives:

$$\boxed{B_{\mathrm{EGF}}(z) = \sum_{t=0}^{L-1} h_t I^t G(z), \qquad IG(z) = \int_0^z G(w)\,dw.}$$

Proof. The EGF of the delayed sequence is $\sum_{m\geq t} a_{m-t} z^m/m! = I^t G(z)$. Summing taps gives the formula. Conversely, $G^{(t)}$ has coefficient sequence a_{m+t}, so derivatives sample future coefficients and do not implement these past delays. □

For two EGFs G_a, G_h, ordinary function multiplication instead gives the **Hurwitz product**

$$G_a(z)G_h(z) = \sum_m \left[\sum_{t=0}^{m} \binom{m}{t} h_t a_{m-t}\right] \frac{z^m}{m!}.$$

This is a binomial convolution. For example, with $a_m = 1$ and taps $h = (1,1)$, ordinary taps give $b_2 = 2$, while the Hurwitz product gives $b_2 = 3$. This explicit negative control catches a common normalization error.

Intrinsic nonlinear feedback taps are the parameters (h_t, q_t, r_t) inside $F' = 1 + \sum_t h_t F(q_t z)F(r_t z)$. They change the function's defining equation and its derivative recurrence. Ordinary FIR taps are a linear transformation of an already defined coefficient sequence. Both are useful, but they operate on different objects. The intrinsic-bank growth theorem belongs to `routes/asymmetric_dilation.md`.

E.5 The exact finite polyphase tap matrix

For an OGF sequence a_m, define lanes $x_l[n] = a_{bn+l}$ and $X_l(w) = \sum_n x_l[n] w^n$. Apply finite ordinary taps and obtain $y_j[n] = b_{bn+j}$. Then the exact matrix identity is

$$\boxed{Y_j(w) = \sum_{l=0}^{b-1} H_{jl}(w) X_l(w), \qquad H_{jl}(w) = \sum_{\substack{0\leq t<L \\ t+l\equiv j\ (b)}} h_t w^{(t+l-j)/b}.}$$

Every exponent is a nonnegative integer. It records the carry across phase boundaries, rather than erasing it.

Proof. A term $h_t a_{bn+j-t}$ is in phase l exactly when $t + l \equiv j \pmod b$. Write $d = (t+l-j)/b$. That term is $h_t x_l[n-d]$, whose OGF is $h_t w^d X_l(w)$ by zero extension. Summing over taps and phases gives the matrix. □

For two phases and two taps,

$$\begin{pmatrix} Y_0 \\ Y_1 \end{pmatrix} = \begin{pmatrix} h_0 & h_1 w \\ h_1 & h_0 \end{pmatrix} \begin{pmatrix} X_0 \\ X_1 \end{pmatrix}.$$

The upper-right factor w is necessary: a delay from odd phase to even phase crosses one block. A diagonal-only phase operation cannot implement this filter. The code constructs the sparse matrix for any finite base and tap vector and checks that recombination gives the full direct convolution.

E.6 Reuse and evidence

From the project directory, standard Python is enough:

```
python family_lab.py coefficients --q 1/4 --r 3/4 --n 8
python family_lab.py catalog --n 12 --output-dir examples
python checks/family_lab_checks.py
```

The reusable functions include `asymmetric_odd_derivatives`, `feedback_odd_derivatives`, `asymmetric_polynomials`, `section`, `phase_lanes`, `merge_lanes`, `ordinary_taps`, `hurwitz_product`, `derivative_shift`, `integral_shift`, `polyphase_tap_matrix`, and `apply_polyphase_matrix`. All core arithmetic is `fractions.Fraction` or integer arithmetic. The phase and tap API applies to any finite rational sequence, not just the special-function catalog.

The final completed run passed **8,038 exact assertions** and is recorded in `evidence/family_lab_checks.json`. Its scope includes:

- 18 catalog rows through A_{12}, independently compared with dense formal Picard iteration through degree 25.
- Bivariate polynomials through A_8, with 540 nonzero monomials checked for integer positivity, symmetry, parity and the degree bound.
- Tangent coefficients independently obtained by division of sine and cosine, plus hyperbolic tangent, scaled sine and linear anchors.
- Root-filter multipliers reduced exactly in cyclotomic fields for bases one through seven and indices zero through 31.
- Direct FIR convolution compared with phase-matrix convolution for 26 input lengths, seven bases and four tap vectors, including signed rational taps.
- The antiderivative interpretation of EGF taps and the factorial conversion of Hurwitz products, independently compared with ordinary Taylor products.
- Negative controls for omitted phase carries, aliasing lanes by concatenation, silent truncation, derivative/delay confusion and Hurwitz/FIR confusion.

The bounded checks validate the implemented paths in the recorded ranges. The universal polynomial, degree, clearing, root-filter and phase-matrix statements follow from the proofs above. The finite checks do not establish entire continuation, growth order, or historical novelty. Those claims are separately proved or explicitly left unresolved in the project.

`evidence/family_lab_checks.json` records software version, code hashes, exact arithmetic, bounds, category counts and run time. It reports no failed assertion in this run. Independent audit identified two API input-validation defects: a fractional residue was accepted by the section operator, and an empty phase matrix accepted base zero. Both are now rejected and have passing regression checks. These defects affected invalid inputs, not the displayed identities. A draft prose example initially transcribed $A_3(1,2)$ and $A_4(1,2)$ incorrectly; the exact catalog comparison exposed the mismatch and corrected them to 35200 and 38041600. The code result did not change. Larger numerical evidence is unnecessary for these algebraic identities; an independent mathematical audit should examine the reductions.